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Abstract

London Fire Brigade publishes detailed incident and mobilisation records, but the public data are difficult to interpret as evidence for response-performance questions: borough demand, incident type, response-time denominators, appliance-level mobilisation, and spatial uncertainty all interact. This project asks how an uncertainty-aware visual analytics dashboard can support structured inspection of London Fire Brigade incident demand and first-pump response performance using public open data.

The project builds a reproducible Python data pipeline over the 2024–2026 LFB Incident Records, LFB Mobilisation Records, London borough boundaries, and a public fire-station address/postcode table. The processed artefacts feed a Flask API and a D3.js linked-view dashboard with borough map, monthly trend, response-time distribution, and ranking-table panels. The dashboard is deliberately framed as an exploratory evidence surface rather than an operational dispatch, routing, forecasting, optimisation, or station-relocation system. Data-quality constraints are carried through the pipeline and report: only 37.9% of incident rows have precise coordinates, mobilisation-derived claims are filtered to matched canonical incidents, and the station layer distinguishes incident-derived assignment footprints from postcode-geocoded public station-address proximity.

The assembled evidence shows that the dashboard can expose borough-level response-performance patterns and the caveats needed to interpret them. Across incidents with recorded first-pump attendance time, 32.4% exceed six minutes; the highest exceedance shares are in Hillingdon (52.2%), Bromley (47.6%), and Richmond upon Thames (45.2%), and the ranking remains stable when false alarms are excluded. Precise-coordinate availability varies strongly by borough, so street-level density views are treated as diagnostic rather than as complete demand maps. Mobilisation matching retains 438,211 rows for deployment-side qualification, and the public station-address recovery quantifies the difference between assignment footprints and station-address proximity rather than treating them as the same thing. The evaluation combines a heuristic review, task-based inspection, and a compact think-aloud summary focused on comparison behaviour, uncertainty noticing, and station-coverage overclaim risk. The contribution is therefore an auditable visual-analytics artefact and evidence pipeline, plus an explicit account of what the open data can and cannot support.

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1 Introduction

1.1 Background and Motivation

Urban emergency response is a core problem for urban informatics: public agencies must allocate scarce resources across unequal neighbourhoods while explaining performance in a form that planners, researchers, and residents can inspect. Open-data portals improve transparency, but denominators, missing values, spatial disclosure control, and operational caveats can still make a simple map look more authoritative than the evidence allows. London Fire Brigade (LFB) provides a useful case because it publishes detailed incident records, yet public-facing analysis remains limited to static dashboards and tabular reports. The 2024–2026 slice used here contains 293,646 incidents over 26 months, roughly 135,000 per year across 33 boroughs, from fires and medical co-responses to special-service calls and false alarms. This creates a specific visual-analytics gap: how to inspect response-performance patterns while keeping uncertainty visible.

Initial exploratory analysis of the most recent 26 months of LFB incident data (January 2024–February 2026, 293,646 records) confirms three motivating problems for this project:

- **A consistent service-equity gap.** 32.4% of incidents with recorded first-pump attendance time exceed the six-minute target. The gap is geographically structured: the highest recorded-time exceedance shares are in outer-London boroughs including Hillingdon (52.2%), Bromley (47.6%), and Richmond upon Thames (45.2%), while several inner-London boroughs sit substantially below the London-wide reference line.
- **A resource-utilisation question.** False alarms account for 46.5% of all incidents and consume more pumps per incident on average (1.9) than fires (1.7). The cost-versus-benefit of dispatching to false alarms is not surfaced in current LFB public reporting.
- **A spatial data-quality problem.** Only 37.9% of records carry precise coordinates; LFB suppresses the rest on disclosure-control grounds, releasing them at 100 m-grid resolution only (§3.3). Street-level spatial analysis must acknowledge this constraint, and it motivates a visual analytics design that surfaces uncertainty rather than hiding it.

These motivating numbers are summarised in Table 1; the data conventions behind them are stated in §1.4.

Metric	Value	Interpretation
Total incidents (Jan 2024 – Feb 2026)	293,646	Dataset scope
First-pump attendance > 6 minutes	32.4%	Service-equity motivation
False alarms (share of all incidents)	46.5%	Resource-utilisation question
Records with precise (non-suppressed) coordinates	37.9%	Spatial-analysis limitation

Table 1: Motivating descriptive statistics computed from the project’s data slice (§1.4 specifies field definitions). Each row maps a measured fact to one of the project’s framing concerns.

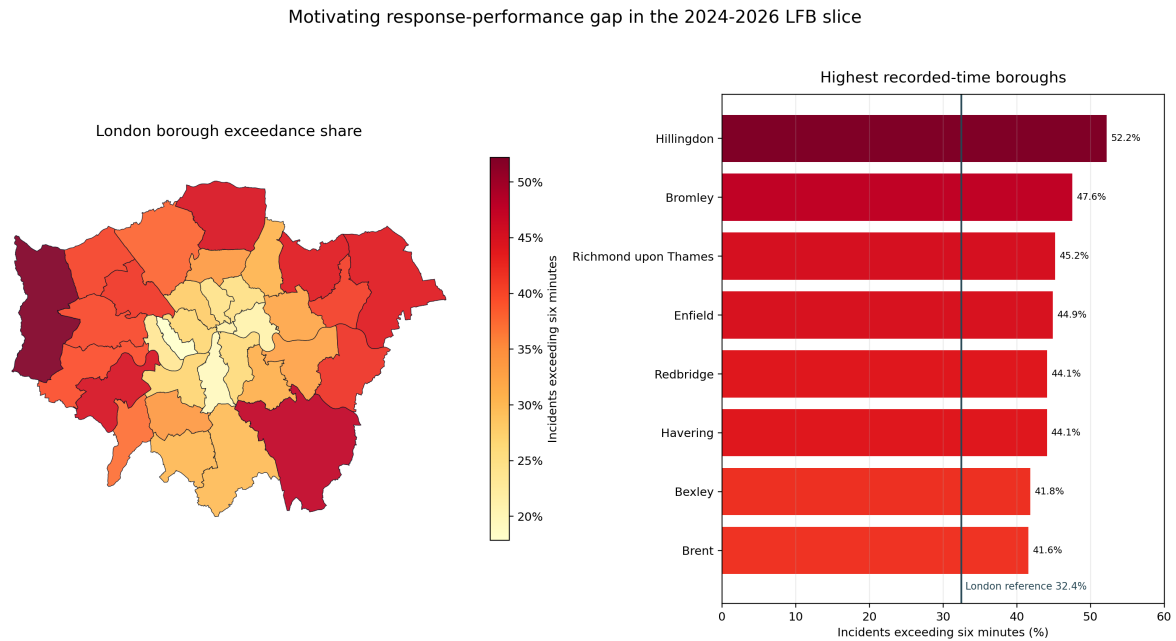


Figure 1: Motivating borough-level response-performance gap in the 2024–2026 LFB slice. The map and bar chart use recorded first-pump attendance times; the vertical reference line marks the project-wide recorded-time exceedance share.

Together with the absence of an interactive open tool for LFB stakeholders, these findings frame the project’s contribution.

1.2 Aims and Objectives

The project aim is to design, build, and evaluate a web-based interactive visual analytics dashboard that helps users ask more defensible questions of LFB open data: where workload and response-time pressure appear together, how strongly those patterns depend on incident type and time, and what caveats are required before a map is interpreted as operational evidence. The dashboard targets two *potential* stakeholder profiles: (a) LFB and Greater London Authority planners exploring borough-level evidence for discussion, and (b) academic and policy researchers studying urban emergency response equity. Operational adoption by either group is outside this project’s scope; evaluation is limited to usability and analytical interpretability rather than operational decision validation (§4.6).

The concrete objectives are:

1. **Data pipeline.** Build a reproducible processing pipeline over LFB incident records, ONS borough boundaries, mobilisation records, and a public fire-station address list (postcode-geocoded for proximity context) that handles the 62% missing-coordinate problem explicitly.
2. **Spatial-temporal analytics.** Implement borough-level aggregations, diagnostic hexbin/density maps for the geocoded subset, and response-time distributions sliced by hour, weekday, borough, and incident type.

3. **Interactive dashboard.** Deliver a Flask + D3.js dashboard with linked-view brushing across map, time-series, distribution, and ranking panels, plus a bounded station-proximity/assignment-footprint evidence strand that is explicitly framed as exploratory context rather than an operational dispatch, routing, coverage, or relocation model.
4. **Evaluation.** Evaluate the artefact against the research questions below using a closed heuristic walk-through, scripted think-aloud/probe instruments, dashboard screenshots, and a compact think-aloud summary.

1.3 Research Questions

The initial project brief listed five example questions. This thesis consolidates them into one primary research question and three supporting questions. The three-question form keeps the brief’s spatial, temporal, visual-analytics, resource-context, and data-quality themes visible while avoiding a scattered evaluation chapter.

1. **Primary RQ.** How can an uncertainty-aware visual analytics dashboard support structured inspection of London Fire Brigade incident demand and first-pump response performance using public open data?
2. **RQ1.** Which boroughs, incident groups, and time periods show persistent incident demand or six-minute first-pump attendance exceedance patterns in the 2024–2026 LFB open-data slice?
3. **RQ2.** How do mobilisation records, false-alarm workload, coordinate suppression, missing response times, and station-address evidence qualify the interpretation of those patterns?
4. **RQ3.** To what extent do the linked dashboard and report evidence package help users make useful but qualified analytical judgements without overstating station coverage, routing, or relocation claims?

1.4 Data Sources and Conventions

The primary dataset is the **LFB Incident Records** open dataset [1] published on the London Datastore, used here as the 2024-01 to 2026-02 slice (293,646 rows $\times$ 39 columns, 63 MB). The implementation also uses the ONS / GLA borough boundary layer [2], LFB mobilisation records [3], incident-derived borough and station footprint centroids, and a London Datastore public fire-station address list [4] whose postcodes are geocoded for a bounded station-proximity check in §5.6. The address list is not treated as a routing-grade station entrance dataset. All currently-used data are open and require no institutional access.

The descriptive statistics quoted in §1.1 were computed directly from this slice. Specifically: response-time exceedance is defined as the `FirstPumpArriving_AttendanceTime` field exceeding 360 seconds (the 6-minute LFB attendance benchmark [5]); “precise coordinates” refers to records with non-null `Easting_m` / `Northing_m` fields within the Greater London bounding box; the 100 m-grid `Easting_rounded` / `Northing_rounded` fields are present for essentially all records (§3.3); borough medians exclude null and out-of-range entries. Full notebook reproducing these

figures: `notebooks/01_initial_eda_executed.ipynb`. The 62% suppressed-coordinate share is a known constraint: any spatial-density or station-proximity analysis based on the precise 37.9% may be biased if suppression is not random across borough, incident type, or time-of-day. Borough- and grid-level work in this report therefore uses the near-complete rounded family, and the dashboard surfaces per-borough precise-coordinate coverage rather than assuming suppression is at random.

1.5 Methodology Overview

The reported work followed a four-phase workflow:

- **Phase 1** (April 2026, completed): data acquisition, exploratory analysis, literature review; closed at the Preliminary Report submission on 30 April 2026.
- **Phase 2** (May–June 2026, completed): dashboard implementation in Flask + D3.js with iterative internal review.
- **Phase 3** (July 2026, completed for this report): evaluation, usability testing, and performance profiling.
- **Phase 4** (post-report): presentation and any further code-release housekeeping outside the evidence assessed in this report.

Tooling: Python 3.12 (`pandas`, `GeoPandas`, `PySAL`) for the analytic backend, D3.js v7 [6] for client-side visualisation (selected for its representation-transparent binding of data to DOM elements, which makes the linked-view pattern straightforward to implement), Flask for the API layer, Git for version control, and Zotero for reference management.

1.6 Report Structure

The remainder of this report is organised as follows. §2 reviews the relevant literature and synthesises three cross-strand patterns that frame the project’s contribution. §3 supplies the technical and domain background assumed by later chapters. §4 specifies the dashboard’s design and design rationale, mapping each choice to the literature positions of §2. §5 records the implementation as it is built. §6 reports evaluation results against the research questions of §1.3. §7 discusses legal, social, ethical, and professional considerations. §8 concludes with research-question answers, contribution statement, future work, and lessons learned. Appendix A indexes source code and reproducibility commands, Appendix B lists data, figure, and evaluation artefacts, and Appendix C provides the dashboard user guide.

2 Literature Review

2.1 Scope and Method

A targeted search of Google Scholar, IEEE Xplore, MDPI, and ScienceDirect was performed across six themes – visual-analytics design and evaluation, visual analytics for emergency management, spatial analysis of incident data, fire-station optimisation, London-specific work, and predictive analytics for emergency events. Thirty papers spanning 1974–2024 were read in depth; the review distinguishes foundational design references (Shneiderman’s mantra; Munzner’s nested model; Church and ReVelle’s covering-location formulation) from contemporaneous comparable work (Park et al. on Seoul, Bispo et al. on Aveiro, Taylor on London 2012–2014) and from the uncertainty-visualisation, modern-dashboard, and distributive-justice literatures that frame the dashboard’s core design decisions. Each cited paper was checked against its primary source where the full text was accessible, rather than against secondary summaries; for older foundational works where full text was not accessible, the relevant formulations were cross-checked against publisher metadata and subsequent peer-reviewed uses. Where common framings of the literature diverge from the actual published work, the prose below reflects the latter.

2.2 Visual Analytics: Design Principles, Tools, and Emergency Applications

Dashboard design for spatial-event data follows Shneiderman’s Visual Information Seeking Mantra – “overview first, zoom and filter, then details-on-demand” [7] – operationalised through a task-by-data-type taxonomy of seven tasks (Overview, Zoom, Filter, Details-on-demand, Relate, History, Extract). The dashboard’s interaction repertoire is further organised around the user-intent categories of Yi et al. [8]: Filter (borough/year/category selection), Connect (brushing across linked map and time-series), Explore (map pan/zoom), and Abstract/Elaborate (hover tooltips). At the implementation layer, the dashboard adopts the declarative-visualisation paradigm [9, 6], which separates visual specification from execution by binding data to W3C DOM elements (SVG/HTML) via the `enter/update/exit` pattern; D3.js [6] is the web-native realisation of that paradigm. Munzner’s nested design model [10], elaborated in book-length form in *Visualization Analysis and Design* [11], provides the cross-cutting validation framework: characterise task and data first, then choose operations and data types, then design encodings and interactions, then implement – because the model gives an explicit place to record decisions at each layer and makes it harder to mask a poorly-defined task with good-looking visuals. Section 4.6 utilizes the seven-scenario evaluation taxonomy of Lam et al. [12] to organize the evaluation strands. This taxonomy surveys 850 InfoVis papers and consolidates evaluation goals into scenarios spanning user experience, performance, and visual data reasoning.

Where Shneiderman’s mantra frames an analyst’s interaction with arbitrary data, more recent work argues that *dashboards* form a distinct sub-class with their own design space. Sarikaya et al. [13] structure that space along four axes – purpose, audience, visual-and-interactive features, and data semantics – and derive seven dashboard genres from a corpus of 83 in-the-wild examples. Bach et al. [14] extend the corpus to 144 dashboards and consolidate 42 design

patterns across content, composition, and interaction, articulating a four-parameter trade-off between abstraction, screenspace, pagination, and interaction (any reduction along one parameter forces an increase along another). The proposed LFB dashboard sits at the intersection of the operational-decision-making and public-communication genres of [13], and instantiates the analytic genre of [14] with a stratified page layout, hierarchical drill-down (London $\rightarrow$ borough $\rightarrow$ station-related context), and a schematic-spatial layout for comparing linked borough evidence.

Within emergency management specifically, a 2016 systematic mapping by Dusse et al. [15], complemented by the visualisation-focused mapping of Marbouti et al. [16], established that maps and dashboard composites dominate the literature, but most reviewed systems are post-hoc analytical tools rather than operational decision-support interfaces. Reinert et al.’s PanViz 2.0 [17], the closest published architectural exemplar, is a Flask-backend, React-frontend visual-analytics web application for US county-level pandemic decision support. Its deployed system uses a three-panel layout (parameter control, spatial choropleth, temporal trend), and the paper additionally describes a planned next-generation what-if mode in which users perturb intervention parameters and visually compare scenarios. The deployed three-panel layout is what the present project adapts directly, with two adaptations: the geographic granularity moves from US county to London borough, and the frontend stack moves from React to D3.js, chosen for the data-to-DOM binding that makes the linked-view brushing pattern explicit in the source. The principal methodological difference is that PanViz 2.0 is forecast-oriented (calibrating a closed-form epidemic model with observed data) while the present project is descriptive (showing observed LFB response patterns and the data’s uncertainty), so the analogy is at the dashboard-architecture level, not the analytical-task level.

Beyond these exemplars, alternative input modalities have been explored: Imran et al.’s survey [18] catalogues social-media-driven situation awareness during mass emergencies, and AR-augmented spatial communication [19] extends the modality further. These alternative streams are noted for completeness; their applicability to a planning-focused LFB dashboard, which works from structured incident records rather than user-generated content, is peripheral.

2.3 Uncertainty-Aware Visualisation

The project’s framing as an *uncertainty-aware* dashboard is grounded in three complementary literatures, none of which has previously been brought to bear on emergency-services performance data. From the visualisation-design side, Hullman, Resnick and Adar [20] show in a 288-subject controlled experiment that static error bars and violin plots produce mean absolute errors three-to-four times larger than animated Hypothetical Outcome Plots on probabilistic-comparison tasks (estimating $\Pr(B > A)$ across pairs of distributions); reliance on point estimates and binary-coverage error bars therefore systematically misleads non-expert readers, the precise audience for a public-facing performance dashboard. From the cartographic side, MacEachren et al. [21] establish empirically that only *fuzziness*, *location offset*, and *colour value* are intuitively read as uncertainty by GIScience-literate users, while colour saturation – the variable most commonly assumed to encode confidence – is not; encoding the LFB dataset’s geocoding incompleteness (only 37.9% of records carry valid coordinates) therefore demands

fuzziness or value ramps rather than the saturation channel default. From the cognitive-science side, Padilla, Creem-Regehr, Hegarty and Stefanucci [22] embed uncertainty visualisation in dual-process decision theory: visual-spatial biases such as the deterministic-construal error (e.g. misreading error bars as absolute high/low bounds, or reading the Cone of Uncertainty as physical growth) arise during Type-1 bottom-up attention and resist correction even when a key is provided. The implication is that the LFB dashboard cannot rely on legends or footnotes to discharge its uncertainty-awareness; the encodings themselves must be Type-1-readable. The current implementation enacts a narrower first iteration: median-to-p95 ranges, explicit denominator text, and precise-coordinate coverage fields. Full quantile dotplots, map fuzziness, and natural-frequency tooltip phrasing remain second-iteration items rather than completed results.

2.4 Spatial Analysis of Emergency and Fire Incidents

Seoul grid evidence: The methodologically closest comparable study at comparable scale is Park, Kwon and Azam’s GIS-based spatial analysis of Seoul fire-emergency response [23]. After filtering 846,384 raw 2017–2022 Seoul rescue reports, they retain 56,514 life-saving cases on a 500 m × 500 m grid covering Seoul (3,227 grids). The analytic threshold is 5 minutes from dispatch to scene, used as the operational proxy for the broader “golden time” concept. They report Global Moran’s $I = 0.45$ ($p < 0.001$) for the full incident distribution and 0.57 for the >5-minute subset, and use Local Moran’s I (LISA) to localise hotspots. Three findings inform this project: (i) the >5-minute subset is more spatially clustered than the full incident distribution, indicating that response equity is geographically structured beyond what raw incident density would predict; (ii) the Seoul mean grid-to-nearest-station straight-line distance of 1.34 km masks district-level variation, with Songpa-gu, Gwanak-gu and the area south of the Han River exceeding 1.45 km, and traffic volume alone does not explain the >5-minute clustering, attributing it instead to station-spacing gaps; (iii) the proxy of “nearest-station” distance is computed straight-line, which supports using transparent straight-line station-address proximity as context while still stopping short of a network-routing model. The 5-minute Seoul threshold is the methodological analogue of the LFB 6-minute first-pump target. Two limitations of Park et al. shape the present design: their grid-level coverage analysis depends on geocoded arrival points (LFB has only 37.9% geocoded), and their proximal-station definition uses straight-line rather than road-network distance.

Taylor closure cohort: Taylor’s spatial-survival analysis of LFB dwelling-fire response times [24] fits a parametric proportional-hazards model on 2012, 2013, and 2014 separately, with harmonic regression terms (periods 1, 1/2, 1/3, 1/4 day) for diurnal effects and shared spatially-correlated frailties on a 500 m grid for residual spatial structure. Two findings ground the present project. First, time-of-day effects are invisible in raw aggregation but emerge after spatial adjustment: 3am–7am and 11am–6pm relative risk falls below 1 (slower response), with the trough at 5am at ≈ 0.7 . The dashboard’s time-series panel must therefore offer both raw and spatially-adjusted views, otherwise users see a flat trend and infer there is none. Second, of the ten London stations closed in 2014, eight (Belsize, Downham, Kingsland, Knightsbridge, Silvertown, Southwark, Westminster, Woolwich) had surrounding 500 m squares with $\geq 80\%$ posterior probability that 2014 $P(\text{response} > 360\text{s})$ was at least 0.1 higher than in both 2013 and

2012; Bow and Clerkenwell were not similarly affected. This is the reference cohort against which the 2024–2026 borough patterns can be cross-checked, with the explicit caveats that (i) Taylor analyses dwelling fires only and (ii) Taylor’s 500 m granularity is finer than the borough-level views feasible from the 37.9% geocoded share of the present project’s data.

Coverage models as boundary evidence: Service-coverage modelling for fire risk in mid-sized cities [25] demonstrates that isochrone-based reachability analysis at multiple cut-off thresholds (3, 5, 10 minutes) can surface response-window gaps when facility locations, hazard sites, and travel-time assumptions are available. For this project, the value of that literature is mainly boundary-setting: it shows what a routing-grade coverage surface would require, while the present open-data artefact can only provide public station-address proximity and assignment-footprint context. The isochrone-surface concept itself is formalised by Andrienko and Andrienko [26] as “a series of concentric polygons centred on a selected location, representing the areas accessible from this location within specified time budgets (e.g. 3, 6, 9 minutes)” – a definition that maps conceptually onto the LFB six-minute first-pump target but is not implemented here. Potential spatial accessibility is commonly operationalised by the two-step floating catchment area (2SFCA) method [27], which combines a supplier-side service-area ratio with a demand-side aggregation under a single travel-time threshold; Luo and Wang further show that 2SFCA is a special case of the gravity model with a dichotomous travel-time impedance, exposing the artefact of binary thresholds analogous to the LFB six-minute target. 2SFCA sits within a broader landscape of accessibility measures classified by Geurs and van Wee [28] along two axes – four components (land-use, transport, temporal, individual) and four measurement perspectives (infrastructure-based, location-based, person-based, utility-based). The LFB dashboard does not implement these accessibility models because open data does not expose routing, capacity, or individual-level demand at the required granularity; instead, it keeps station-address proximity as descriptive context beside borough-level response evidence.

Equity framing: The framing of these spatial gaps as equity questions, rather than purely operational ones, draws on a longer urban-geography tradition. Talen and Anselin’s foundational work on spatial equity [29] demonstrated that the choice of accessibility measure (count, gravity-based potential, average travel distance, distance-to-nearest) materially changes conclusions about whether a public service is equitably distributed – a methodological warning that this project takes seriously by surfacing multiple aggregations (borough median, station-grouped median, hour-of-day breakdown) rather than a single composite KPI. Pereira, Schwanen and Banister [30] update this position with a contemporary review of distributive-justice theories applied to public services: they reject horizontal equality (uniform service everywhere) as a complete equity criterion and adopt a synthesis built on Rawls’ difference principle and Sen’s capability approach, which requires (i) a sufficientarian floor – a socially-defined minimum threshold of accessibility that government must guarantee – combined with (ii) an egalitarian requirement that residual inequalities above the floor be arranged to favour the least-advantaged group. Mapped onto fire-service response, the LFB six-minute first-pump target operationalises the sufficientarian floor, while the borough- and ward-level disparities of attendance time *above* that floor are the egalitarian-layer signal; the dashboard is designed to surface both layers separately rather than collapsing them into a composite KPI. Batty’s commentary on big data

and smart cities [31] argues that real-time and routinely-collected urban data are shifting urban planning’s analytical horizon from years and decades toward the minutes-to-days scale on which disruptions occur, and notes that emergency-services operations research is one of the few domains already operating at this short horizon. The LFB CTRA dataset is event-tagged rather than sensor-streamed, so the analogy is partial; nevertheless, the dashboard’s hour-of-day and day-of-week panels are positioned within this short-horizon framing, treating response-time variability as an operational signal rather than a long-term strategic indicator.

2.5 Fire-Station Placement and Resource Optimisation

Covering-location lineage: A second strand of literature applies optimisation rather than analysis. The covering-location lineage originates with the Maximal Covering Location Problem of Church and ReVelle [32], which formalised the trade-off between number of facilities p and population covered within a service distance S as a 0–1 integer program; alongside the earlier Location Set Covering Problem [33], this is the foundational pairing for emergency-service siting. Operations research models for emergency vehicle deployment have been developed continuously since the mid-1960s, spanning deterministic covering, probabilistic queueing, simulation, and dynamic relocation paradigms [34]; the present project draws on this lineage as visualisation input rather than re-solving the optimisation itself.

Recent optimisation examples: Recent work in this strand combines several of these threads. Bispo et al. [35] provide the methodologically richest example: they model Aveiro fire incidents as an inhomogeneous Poisson point process with a log-cubic predictor and population-density offset; simulate 450 realisations; cluster each via k -means with k matched to the 27 existing fire stations to obtain candidate optimal locations; and define 90%/99% confidence siting regions through bivariate-normal estimation over the simulation set. Voronoi tessellation completes the layout by defining service areas. Bispo et al. explicitly validate the use of Euclidean distance as a road-distance proxy with $r = 0.96$, which helps justify transparent straight-line distance as a limited station-address proximity cue, not as operational coverage evidence. Multi-source geospatial optimisation work for Beijing [36] integrates road network, demographic, and incident-history layers in a multi-criteria objective, and finds the city-level coverage rate falling from 96.51% off-peak to 71.43% at peak hours – a time-varying coverage finding that motivates by-hour inspection while also illustrating why the present project does not claim coverage without a travel-time model. Space-syntax-augmented location-allocation models [37] extend the formulation by chaining maximum-coverage location-allocation with sDNA road-closeness to extend candidate points to buildable regions, motivated by the empirical observation that approximately 60% of fire-department tasks are technical rescues rather than fire suppression – a category imbalance that maps directly onto the LFB’s 46.5%/40.6%/12.9% (false alarm / special service / fire) split. Bolouri et al. [38] extend this strand by combining three objectives – minimising demand-to-station distance, minimising travel time, and maximising five-minute coverage – under explicit station-capacity and ordered-priority constraints, solved by genetic-algorithm and simulated-annealing benchmarks on Tehran’s Region 11.

Boundary for this project: These papers share a common limitation from the present project’s perspective: they produce a static optimisation output (even when, as in Bispo et

al., that output includes confidence regions and multiple strategy classes), not an interactive evidence-inspection surface. The present project deliberately does *not* attempt to solve the location-allocation problem; instead, it provides a visual-analytics surface on which users can compare empirical response-time gaps with station-related context while leaving any optimisation or reconfiguration judgement outside the artefact. This positions the project as complementary to, rather than competing with, the optimisation literature: an external optimisation pipeline could supply candidate layouts for a validated operational study, and the dashboard pattern would then help planners inspect consequences under explicitly stated assumptions.

2.6 LFB-Specific Prior Work

The most directly comparable prior work is the open-source analysis of Yang and Benidir [39], which fits a linear regression of per-borough fire counts against thirteen socio-economic features (population, density, foreigner share, homelessness, life expectancy, education, etc.) with exhaustive feature selection, achieving $R^2 = 0.90$ with a six-variable model. The dominant predictors are population, density, foreigner share, homeless count, life expectancy, and education score – a profile consistent with general urban-deprivation findings. For this project, their results provide a benchmark: the dashboard’s borough rankings should reproduce these correlations qualitatively, and the choice of context layers (population density, IMD) is informed by their variable-selection results. Yang and Benidir’s work is, however, a static modelling exercise without an interactive interface and predates the 2024–2026 data window addressed here, so it does not displace the project’s contribution.

A 2014 GOV.UK analytical blog [40] and an ODI Labs interactive station-closure map [41] are earlier public-facing artefacts that demonstrate continuing public-policy interest in LFB performance, although both use older data and limited interactivity.

2.7 Predictive Analytics

Sharma et al. [42] build Negative-Binomial regression predictors for emergency-event counts at Edmonton’s neighbourhood level (402 areas) and at the fire-station service-area level (30 Voronoi regions), with feature importance pre-selected by Conditional Random Forest. Two findings are relevant. First, predicting total event count is feasible at both spatial levels (neighbourhood weekly MAE 0.81 against an actual mean of 2.06), but prediction of *individual event types at neighbourhood level* is impractical due to zero-inflation, forcing aggregation either to total counts or to coarser fire-station regions – a constraint the present project anticipates given LFB’s 33-borough granularity. Second, post-COVID-19 model errors are heterogeneously affected: medical-event MAE rises around 70%, fire-event MAE rises only around 8%, and motor-vehicle MAE actually declines (post-lockdown traffic reduction). The 2024–2026 LFB window is fully post-COVID, so any historical baseline comparison must allow for category-specific change rather than a uniform pandemic correction. The dominant predictor across most event types is low-income / lower-density-housing share, which echoes the LFB borough-level finding of §2.6 and motivates the choice of IMD as a default contextual layer in the dashboard.

2.8 Synthesis and Identified Gap

Table 2 summarises the seven literature strands against how each is used in this project. Reading across these strands reveals three patterns that no single paper makes visible on its own, and which jointly motivate the present project’s positioning.

Pattern 1: Spatial heterogeneity is universally reported but uncertainty is rarely visualised. Park et al. [23], Taylor [24], Bispo et al. [35], Wang et al. [36], Tian et al. [37], and Bolouri et al. [38] all surface spatial heterogeneity in incident density or response time at sub-city granularity. Yet none of these papers *visualises* the uncertainty in its estimates: Taylor’s 80% posterior probability is rendered as a binary “concerning” tag rather than a continuous credibility surface; Bispo et al. produce 90%/99% siting-confidence ellipses but a single static map; Park et al. render Moran’s I as a single point estimate without dispersion. The literatures Hullman et al. [20], MacEachren et al. [21] and Padilla et al. [22] diagnose for general InfoVis – that point-estimate displays systematically mislead non-expert decision-makers – have not yet been applied to fire-service performance data. The present dashboard and report make a first, bounded attempt to close that gap through median-to-p95 distribution ranges, denominator exposure, precise-coordinate coverage fields, and explicit second-iteration commitments for fuller uncertainty encodings.

Pattern 2: Optimisation outputs are static; decision contexts are scenario-driven. Bispo et al., Wang et al., Bolouri et al., Tian et al., and the OR survey of Goldberg [34] all produce static optimisation outputs – even when, as in Bispo et al., that output includes confidence regions and multiple strategy classes. The actual decision context, however, is inherently interactive: a fire chief weighing a station closure against political feasibility, budget, and public acceptance needs to compare scenarios under their own assumptions, not consume a single “optimal” layout. The dashboard genres of Sarikaya et al. [13] and the design patterns of Bach et al. [14] both name interactive scenario exploration as an under-served pattern; Pereira, Schwanen and Banister [30] make the political-philosophy point complementary: an automated optimum can satisfy aggregate utility while violating distributive-justice constraints that only a deliberative process can specify. The present project deliberately occupies that interactive scenario-exploration role rather than competing on optimisation rigour.

Pattern 3: Sufficientarian thresholds and egalitarian distribution are conflated. Park et al.’s 5-minute Seoul threshold and the LFB’s 6-minute first-pump target are both rendered as *sufficientarian* floors, but the optimisation literature reports binary “coverage” against those floors and rarely discusses within-threshold variance. Pereira et al. [30] argue, drawing on Rawls and Sen, that distributive justice in public services demands *both* a sufficientarian floor *and* egalitarian distribution above it; horizontal equality (uniform service) is a category-coarse target that can mask group-level disparities. No reviewed paper combines both layers of evaluation. The design decision to surface “% of incidents above the 6-minute target” and “borough-level distribution of attendance time below the target” in the dashboard’s linked panels, rather than as a single ranking, operationalises that two-layer view directly.

Two methodological choices warrant justification. First, visual analytics was selected over predictive modelling along the lines of [42] because the intended user is exploring where response-time gaps are and where the data is too incomplete to trust, not forecasting next month’s incident

Strand	Method	How used here
VA design & dashboards	Mantra; nested model; pattern catalogues	Dashboard interaction structure; design-language anchor; WP4 plan
Uncertainty visualisation	HOPs; visual-semiotics; cognitive framework	Median-p95 ranges and coverage exposure now; quantile/fuzziness/natural-frequency encodings as second-iteration targets
VA for emergency mgmt	Linked map / time views	Three-pane layout, linked brushing
Spatial response-time modelling	Hazard models on LFB data	Equity framing; 2014 closure anchor
Location-allocation optimisation	Mixed-integer / point-process	Informs the proxy; not an optimisation engine
LFB-specific prior work	Static regression, blog dashboards	Borough benchmark for descriptive views
Spatial equity / smart cities	Accessibility; Rawls + capability approach	Sufficientarian floor + egalitarian layer; short-horizon framing
Predictive analytics	NB regression + CRF	Justifies VA-not-prediction choice; post-COVID baseline caveat

Table 2: Literature strands and how each informs this project.

count; Hullman et al. [20] document how single-point predictions are systematically mis-read as deterministic. Second, exploratory station-related context was selected over location-allocation optimisation as in [38, 36, 35] because a real optimisation needs routing, dispatch rules, and station capacity, none of which LFB open data exposes at the right granularity. A black-box “optimal layout” would too easily be read as a relocation recommendation, which the station-context framing and the distributive-justice constraints articulated by [30] both argue against.

The project therefore focuses on a practical problem in the LFB data: how to let users explore borough-level response-time gaps while still seeing where the spatial evidence is incomplete. It builds this as an **uncertainty-aware visual-analytics interface** for emergency-service response equity. The linked-view layout is drawn from PanViz, the borough-level response framing from Park et al., the 2014 closure-cohort comparison from [24], and the two-layer equity-evaluation logic from [30]. The 62% missing-coordinate share is treated as something the user should see in the interface and report-side evidence, rather than as a preprocessing assumption to be hidden. The 2014 closure cohort is used as a historical anchor for response-time geography, not as a claim that those closures alone explain what we see in 2024–2026; intervening changes in population, traffic, and service organisation are acknowledged.

3 Background

This chapter provides the technical and domain background needed to read the rest of the report. The literature review (§2) covers research positioning; this chapter covers the standing concepts and dataset specifics that the design and implementation chapters assume.

3.1 The London Fire Brigade and the 2024–2026 Open Data Window

The London Fire Brigade (LFB) is the statutory fire and rescue authority for Greater London, operating 102 fire stations (the count of distinct `IncidentStationGround` values present in the 2024–2026 dataset) across the 33 London local-authority areas (the 32 London boroughs plus the City of London Corporation) and responding to approximately 135,000 incidents per year (293,646 incidents observed over the 26-month slice used here). Its operational performance is published against a headline first-pump attendance target: the first pump should attend the scene within **six minutes** (360 seconds) of the call being received [5]. Throughout this report, response-performance shares are computed over *recorded* attendance times only: for any subset A of the canonical incident slice,

$$E(A) = \frac{|\{i \in A \cap R : t_i > 360 \text{ s}\}|}{|A \cap R|}, \quad R = \{i : t_i \text{ recorded}\}, \quad (3.1)$$

where t_i is the first-pump attendance time of incident i in seconds. Every exceedance share quoted later — from the slice-wide 32.4% of §1.4 to the per-cell dashboard values of §5.2 — is an instance of $E(A)$ for a stated subset A . The LFB Incident Records dataset [1], published continuously on the London Datastore under the Open Government Licence, releases one row per incident with 39 fields covering call, dispatch, attendance times, incident classification, and partial location information; this project uses the 2024-01 to 2026-02 slice (293,646 rows, 26 calendar months, fully post-COVID).

Three features of the dataset shape every downstream design choice in this report. First, incidents are classified into a top-level three-way taxonomy — *Fire* (12.9%), *False Alarm* (46.5%), and *Special Service* (40.6%, including medical co-responses, road traffic collisions, persons-trapped, and flooding) — so the 6-minute target is averaged across a workload that is dominantly non-fire. Second, the dataset records what each incident *cost* in resources (`NumPumpsAttending`, `NumStationsWithPumpsAttending`) but not the dispatch *rule* that produced that cost: per-station capacity, ride-time policies, and call-prioritisation logic are operational data not released publicly. Third, the dataset is event-resolution but not mobilisation-resolution: a sustained fire that returned multiple appliances over an hour appears as a single row, not as a mobilisation timeline. These three features bound the analytical questions the dashboard can support; the design (§4) and implementation (§5) chapters return to each constraint explicitly.

3.2 Visual Analytics: Operating Definitions

The literature review (§2) cites several adjacent communities — information-visualisation, geo-visualisation, dashboard research, uncertainty visualisation, cognitive science of decision-making — whose vocabularies sometimes use the same word for slightly different things. To avoid

ambiguity later, this report adopts the following operating definitions, each anchored to one of the §2 references:

- **Visual analytics (VA)** — following the Thomas & Cook 2005 framing carried by Andrienko & Andrienko [26] and operationalised by Munzner [10, 11], the synergetic combination of database operations, computational analysis, and interactive visualisation in service of a human analyst’s reasoning task. The emphasis is on *interactive reasoning support*, not on producing a static report.
- **Dashboard** — following Sarikaya et al. [13] and Bach et al. [14], a multi-view interactive visual display whose layout, interactivity, and data semantics are bounded by an explicit purpose, audience, and update cadence. “Dashboard” here is therefore narrower than “visualisation”: a single map is a visualisation but not a dashboard.
- **Linked view / brushing** — following Yi et al.’s *Connect* interaction category [8], a multi-panel visualisation where a selection in one panel propagates highlight to the corresponding records in every other panel. The dashboard’s borough → time-series → distribution propagation is the canonical instance.
- **Uncertainty encoding** — following MacEachren et al. [21], the use of a visual variable or adjacent exposure field to make data uncertainty visible *at the same level of attention* as the data point itself. In the implemented first iteration, this is realised through denominator exposure, median-to-p95 response ranges, and precise-coordinate coverage fields; fuller quantile dotplots, fuzziness, and natural-frequency tooltips remain second-iteration items after Hullman et al. [20] and Padilla et al. [22].
- **Station-related context** — this project’s term for the exploratory assignment-footprint and public station-address proximity layer of §4.5. It is a straight-line context cue around historical incident responsibility and postcode-geocoded station addresses, not a routing-grade station-coverage, dispatch, or relocation model. The contrast is with the location-allocation optimisation literature surveyed in §2.5 ([32, 34, 35, 36, 38, 37]).

Each definition is referenced verbatim in the chapters that use the term, so a reader can resolve any later ambiguity by returning to this subsection.

3.3 Geospatial and Temporal Granularities

The dashboard works at three nested geospatial granularities, all derived from the same input record. Greater London is divided by the Office for National Statistics into 33 *boroughs* (the political unit), each containing on average 18–20 *wards* (the administrative-management unit), each in turn covering one or more *Lower-Layer Super Output Areas* (LSOAs, the statistical-disclosure unit, mean population $\approx 1,500$). The LFB Incident Records release exposes each of these via dedicated columns (`IncGeo_BoroughName`, `IncGeo_WardName`, and the LSOA-derived `Easting_rounded` / `Northing_rounded` pair); the 2024 vintage of the ONS Open Geography Portal boundaries [2] is used for joins.

A subtler feature of the schema is that the dataset releases *two coordinate families* that must not be conflated. The precise per-incident coordinates (**Easting_m**, **Northing_m**, plus the derived **Latitude** / **Longitude** and full postcode) are present for only $\approx 37.9\%$ of records: LFB suppresses them on the remaining $\approx 62\%$ on disclosure-control grounds when the address could re-identify an individual. The 100 m-grid coordinates (**Easting_rounded**, **Northing_rounded**) are released for $\approx 100\%$ of records. Borough- and ward-level aggregation can therefore use the rounded family without bias, while diagnostic density/hexbin mapping must use the precise family and explicitly carry the missingness as a 62% gap that is suppressed, and may not be random. The dashboard exposes both as separate dataset facets rather than silently dropping non-precise records (see §5.2). Table 3 generalises this two-family rule into the evidence-support map assumed by §4–§6.

Claim class	Borough / rounded fields	Precise coordinates	Recorded first-pump times	Matched mobilisation partition	Station- address points
Incident demand ranking and monthly trends	Primary	–	–	–	–
Response- performance shares and distributions (Eq. 3.1)	Primary	–	Primary	Qualifier	–
Street-level density diagnostics	–	Diagnostic	–	–	–
Appliance-level deployment patterns	–	–	–	Primary	–
Station-address proximity context	Context	–	–	–	Context
Routing-grade coverage, dispatch, or relocation	Not supported by any family in this evidence base (§4.5)				

Table 3: Evidence-support matrix: which evidence family supports which claim class. Primary carries the claim; Diagnostic supports subset-only robustness checks; Qualifier conditions interpretation; Context is a descriptive cue; a dash marks an unused family.

Temporally the slice covers 2024-01-01 to 2026-02-27 (26 calendar months) and is therefore *entirely post-COVID-19*. Sharma et al. [42] show, on Edmonton EFRS data, that pandemic-era model errors are heterogeneously affected across event types (medical-event MAE rose $\approx 70\%$, fire-event MAE only $\approx 8\%$, motor-vehicle MAE actually declined). Any historical-trend comparison made in §6 against pre-pandemic LFB studies (e.g. Taylor’s 2012–2014 dwelling-fire window [24]) must therefore allow for category-specific level shifts rather than assume a uniform pandemic correction. The dashboard slices time at four granularities — year, month, weekday, and hour-of-day — preserved as derived columns in the canonical dataset described in §5.2.

4 Approach: Specification and Design

This chapter specifies *what the dashboard is designed to do* and *why these design choices have been selected*, in advance of §5 (Implementation), which records *how* the system is constructed.

4.1 Design Goals and Non-Goals

The design pursues three primary goals, each tied to the value-led questions of §1.3:

- 1. **Surface borough-level response-time disparity *with uncertainty***, so that a planner sees not only that borough *X* has a longer median response time than borough *Y*, but also the dispersion within each borough, the incident mix, and the share of incidents whose response time was not recorded. (Serves RQ1–RQ3; encoded after [20, 21, 22] per §4.4.)
- 2. **Provide station-related context without operational overclaiming**, so that assignment-footprint and public station-address proximity evidence can qualify borough comparisons without becoming a routing-grade coverage or relocation model. (Serves RQ2–RQ3; framed as visual-analytics context, not optimisation, per §4.5 and §5.6.)
- 3. **Expose mobilisation and historical context as qualifications**, so that incident-level response summaries can be checked against appliance-level deployment evidence and, where appropriate, interpreted against the 2014 closure cohort identified by [24]. (Serves RQ2.)

The design logic is evidence-led: each research question is tied to a concrete artefact or validation strand, so the report does not drift from available evidence into unsupported operational claims. RQ1 is supported mainly by the borough summary and linked dashboard; RQ2 by the mobilisation matched partition, data-quality sidecars, and station-address recovery; and RQ3 by the implemented linked-view workflow, heuristic walk-through, and prepared think-aloud/probe instruments. Figure 2 summarises this chain and the headline values it carries.

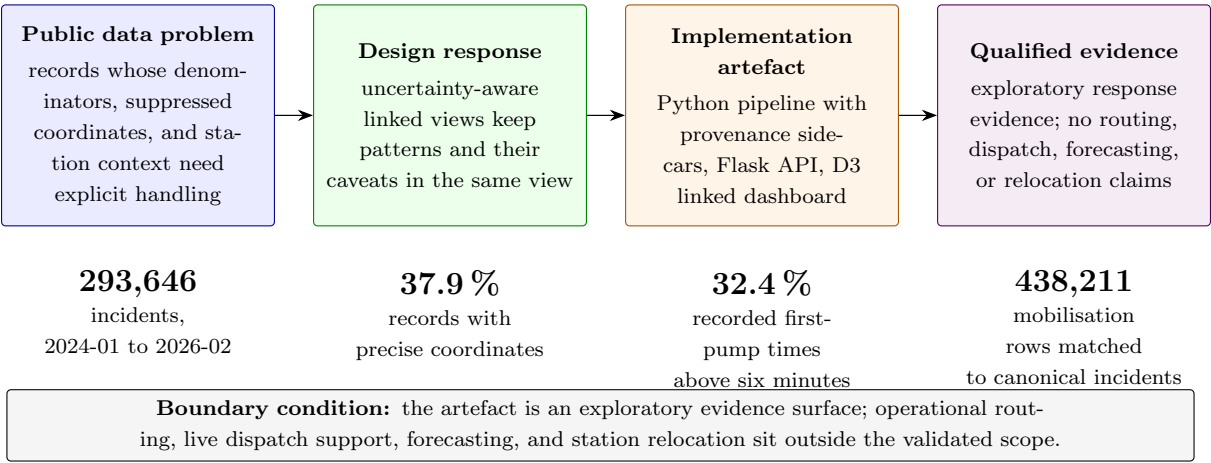


Figure 2: Project evidence map: from open-data problem to qualified visual-analytics evidence. Headline values restate the §1.4 conventions; the boundary strip restates the §4.1 non-goals.

The following are explicit *non-goals*, each ruled out either because the data does not support it (§3.1) or because the literature (§2) shows it would mislead the intended user:

- Location-allocation optimisation. The optimisation literature surveyed in §2.5 produces black-box “optimal layouts” that LFB open data structurally cannot calibrate (no per-station capacity, no dispatch-rule metadata, no routing-graph travel times); see §4.5.
- Real-time dispatch support. The dataset is published with a multi-week lag and the project takes no operational responsibility for live response.
- Predictive forecasting of incident counts. [42] demonstrates that this is feasible at the right scale, but it is a separate research artefact and is left as future work (§8.3).
- Ingestion of social-media or user-generated content. Surveyed by [18, 16], but explicitly out of the project’s scope, which is restricted to structured incident records.

4.1.1 User Scenarios

Three bounded user scenarios drive the requirements. A planning analyst compares boroughs before a review meeting and needs exact denominators after an initial map impression. A policy researcher checks whether a high-exceedance pattern persists after filtering by incident group, month, false-alarm workload, and coordinate coverage. A public-data reviewer tests whether station evidence supports a coverage claim, so the interface must separate assignment footprints from public station-address proximity. These scenarios motivate the linked map, trend, distribution, ranking table, and station-context caveat as one evidence workflow.

4.2 Requirements: Functional and Non-Functional

4.2.1 Functional Requirements

Each requirement maps to one or more research questions of §1.3:

- **F1** (RQ1, RQ3). Render a borough-level choropleth of incident demand and first-pump response performance, with selectable filters for incident group, year, month, weekday, and hour of the day.
- **F2** (RQ1, RQ3). Render a time-series panel of monthly first-pump response time, linked to the choropleth selection via brushing (Yi *Connect*).
- **F3** (RQ1–RQ3). Render a response-time *distribution* panel for the current selection, so that variance and tail behaviour are visible alongside the central tendency.
- **F4** (RQ2–RQ3). Provide station-related context through an assignment-footprint API and a public station-address proximity recovery layer. The implementation must keep these surfaces linguistically separate from routing-grade coverage or relocation evidence.
- **F5** (RQ2–RQ3). Encode coordinate imputation status (precise per-incident versus rounded 100 m grid, per §3.3), missing response-time status, and mobilisation-match status wherever a panel or report-side claim aggregates affected records, after [21].

4.2.2 Non-Functional Requirements

- **N1 Open data only.** The system must run on LFB Incident Records, LFB Mobilisation Records, ONS borough boundaries, and public fire-station address/postcode data ([1, 3, 2, 4, 5]); no KCL Library credentials, no commercial APIs. (See §5.6 D1: the first implementation derives station footprints from the canonical incident table, and the station-address recovery adds public-address postcode proximity without claiming routing-grade coverage.)
- **N2 Single-laptop runnable.** Backend (Flask) and front-end (D3.js) must run on a 16 GB MacBook without GPU, GIS server, or paid cloud service; the canonical parquet (§5.2) is sized so that the entire backing store loads in memory.
- **N3 Browser-rendered.** The user-facing artefact is a web page reachable from any modern desktop browser; no installation, no plugin, no operating-system dependency.
- **N4 Reproducible builds.** Every numeric claim in the report must trace to a script under `src/` or a notebook under `notebooks/`, and every artefact in `data/processed/` must carry a provenance JSON sidecar (see §5.2).
- **N5 Accessible to a non-expert audience.** Following Padilla et al. [22], tooltip and legend wording should make denominator-conditional values readable without specialist knowledge. The first iteration now states that exceedance percentages are among recorded first-pump times; the second iteration should move further toward natural-frequency phrasing (“out of every 100 incidents with a recorded time, X exceed six minutes”) so low-numeracy readers can interpret without working-memory cost.
- **N6 Auditable evaluation.** Evaluation activities (§4.6) must record their participant demographics, task scripts, and observed responses in a form that an external reviewer can reconstruct from the repository.

4.3 Architecture: Three-Pane Linked View

The architecture is a three-pane linked view served from a single Flask process: a borough *map* pane, a temporal *trend* pane, and a response-time *distribution* pane, with a borough ranking table beneath. The three-pane decomposition inherits from the deployed PanViz 2.0 layout [17] (parameter control / spatial choropleth / temporal trend), with two adaptations forced by the LFB context: (i) the geographic granularity moves from US county to London borough, and the temporal granularity from epidemic-day to incident-month; (ii) the front-end stack moves from React to D3.js [6, 9], chosen for the data-to-DOM binding that makes the linked-view brushing pattern explicit in the source rather than hidden inside a virtual-DOM diff.

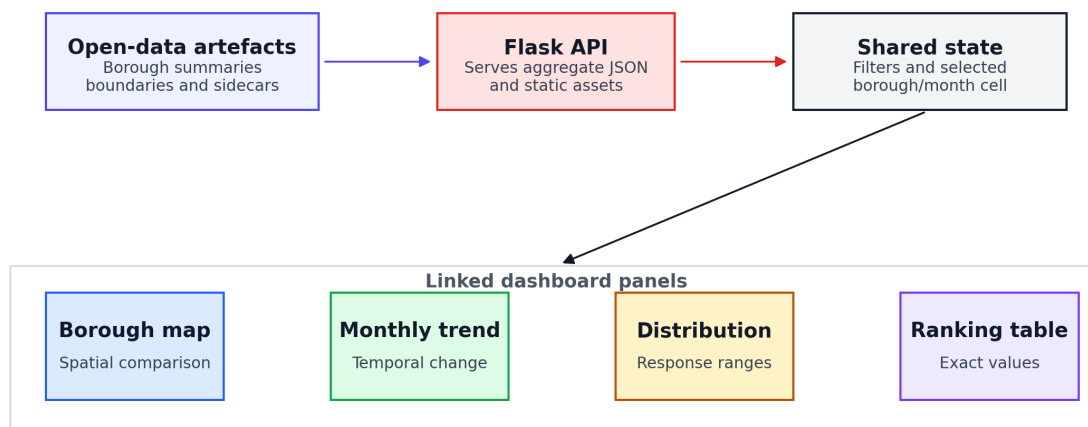


Figure 3: Linked-view dashboard structure. The schematic shows how the filter state and borough selection propagate through the map, monthly trend, response distribution, and ranking-table panels.

The backend / front-end split follows from N1–N3 of §4.2: an open-data, single-laptop, browser-rendered system rules out a separate compute server, but a pure-static deployment would have to ship the canonical 66 MB Excel (or its ≥ 40 MB parquet derivative) to every browser. The adopted hybrid architecture utilizes a Flask process that loads the borough summary parquet once at startup and serves either pre-aggregated JSON or filter-narrowed slices over the same `/api/borough_summary` endpoint; the same Flask process serves the D3 dashboard’s static assets and the borough-boundary GeoJSON from the same origin, which removes the need for cross-origin headers and makes the system executable via a single autonomous Flask instance. The first-iteration front-end issues a single unfiltered fetch at startup (≈ 1.0 MiB JSON, the full 2,574-row aggregate) and applies further filters client-side; an explicit borough-only filter would return ≈ 31 KiB. The endpoint supports both patterns so a future front-end iteration can switch to per-state-change fetches without an API change. The per-incident canonical table is never exposed through the HTTP surface — only aggregations by borough, month, and incident group from §5.2 leave the backend — which is both a privacy posture (no precise coordinates over the wire) and a payload-size posture (1 MiB aggregate JSON, against ≥ 40 MB if the canonical were served whole).

Commercial BI platforms such as Tableau, Power BI, and ArcGIS Online were considered as cheaper implementation paths but rejected for this artefact. The core research object is the uncertainty contract itself: denominator-qualified tooltips, per-cell coordinate-coverage fields, and disclaimer payloads embedded in API responses need to behave as first-class interaction and data-contract rules, not as styling notes attached to a workbook. N1–N3 also require an open-data, no-licence, single-laptop artefact that can be inspected without a paid desktop or hosted GIS workspace. N4 favours script-traceable rebuilds and sidecar provenance over opaque workbook state. Most importantly, a BI skin would hide the encoding-design layer that this

project is evaluating: how linked views, tabular caveat exposure, and station-context wording shape qualified comparison.

4.4 Encoding Choices and Their Justification

Encoding choices follow the §3.2 working definitions, so each visual variable resolves to a single literature anchor rather than an ad hoc designer judgement. The borough-choropleth fill encodes the 6-minute exceedance share via a sequential RGB-basis interpolator (teal $\rightarrow$ amber $\rightarrow$ red), matching the *semantic-colour* dashboard pattern reported as the second-most-common colour treatment in Bach et al.’s 144-dashboard corpus [14] and the operational “traffic-light” convention familiar to fire-service stakeholders. The temporal trend pane renders incident count and exceedance share as two aligned traces sharing a common time axis, supporting the *Connect* category of Yi et al. [8] when a borough is selected on the map. The distribution pane, in its first iteration, renders median-to-p95 ranges with the 6-minute target marked as a dashed reference line; the second iteration replaces this with a full quantile-dotplot rendering after Hullman et al.’s controlled-experiment evidence [20] that static range encodings systematically mislead non-expert readers in probabilistic-comparison tasks.

Uncertainty encoding is operationalised against the empirical visual-variable ranking of MacEachren et al. [21]: the dashboard restricts itself to fuzziness, colour value, and tabular exposure, and explicitly avoids saturation (which the empirical study found to be *not* intuitively read as uncertainty by GIScience-literate users despite its commonplace use). The first-iteration enactment of this contract is the ranking table’s per-borough precise-coordinate-coverage column (the 37.9% / 100% split documented in §3.3 surfaced as a tabular value, not silently hidden) and denominator-qualified hover text for exceedance share; the second iteration adds fuzziness on the map for cells with low precise-coordinate coverage, plus fuller natural-frequency tooltip phrasing after Padilla et al. [22].

4.5 Station-Context Design and the Optimisation Boundary

The station-related layer is an *exploratory context layer*, not a routing-grade dispatch model and not a location-allocation optimiser. The implementation deliberately separates two kinds of evidence: the assignment-footprint proxy of §5.5, derived from historical incident responsibility patterns, and the public station-address postcode proximity recovery of §5.6 D1. Neither is a road-network coverage surface. This boundary follows the optimisation literature: the Maximal Covering Location Problem lineage from Church and ReVelle [32] through contemporary surveys and applications [34, 35, 36, 38] requires per-station capacity, dispatch rules, and travel-time calibration that LFB open data does not expose. Producing an “optimal layout” here would therefore be an unsupported policy recommendation. The project instead uses straight-line distance only as a transparent proximity cue, and treats any routing-grade accessibility or relocation recommendation as future work requiring separate validation. Figure 4 draws the resulting claim boundary explicitly.

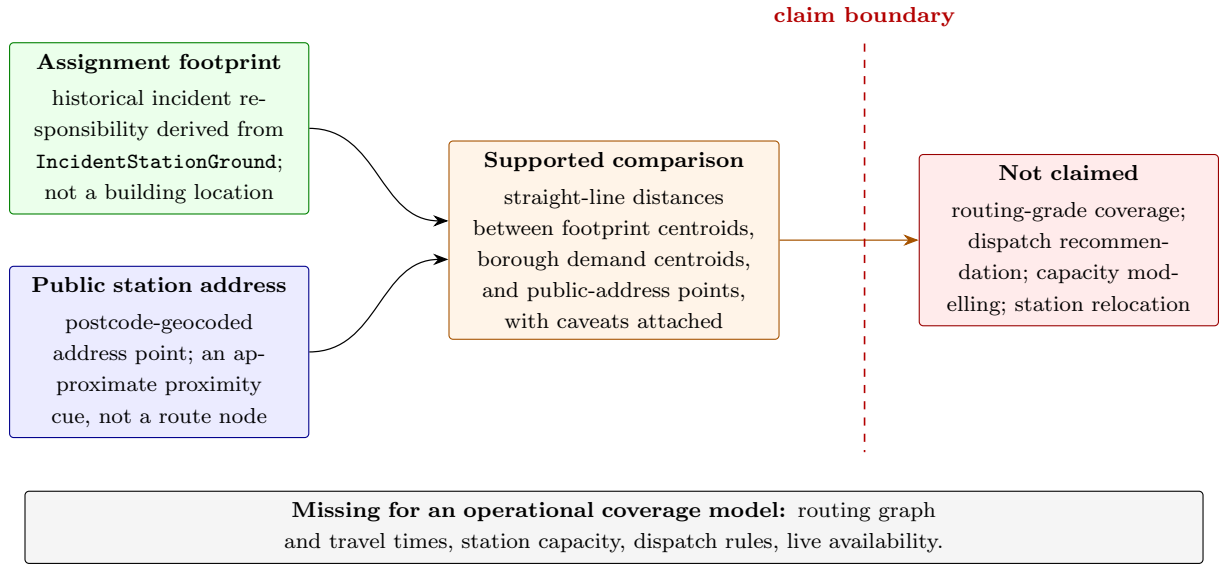


Figure 4: Station-evidence claim boundary. Both station-related layers support straight-line context comparison only; the inputs an operational coverage model requires are absent from the open data, so no such claim is made.

4.6 Evaluation Plan

Evaluation occupies the *Evaluating User Experience (UE)* scenario of Lam et al.’s seven-scenario taxonomy [12], supplemented by light task-based elements where the research questions admit observable behaviour. The plan has three strands. Strand one is a heuristic walk-through using visualisation-adapted heuristics (cluster of *exploration*, *communication*, *abstraction*, and *cognitive fit* concerns from §2.2 and §2.3), feeding the readiness evidence in §6. Strand two is a small-scale think-aloud protocol for three urban-data-domain participants. Strand three is an explicit uncertainty and overclaim probe: whether participants notice and act on coordinate, denominator, mobilisation, and station-footprint caveats, interpreted through Padilla et al.’s dual-process framework [22]. A controlled comparison against a static-baseline dashboard (the UP scenario in its full form) remains out of scope and is named as future work in §8.3.

5 Implementation

This chapter describes how the system specified in the Approach chapter (§4) is built: the codebase layout, the data-engineering pipeline, the front-end visualisation components, the linked-view brushing implementation, and any design-decision deviations encountered during construction.

5.1 System Architecture and Codebase Layout

The repository is laid out around an artefact-first workflow: each named directory holds one class of project artefact — raw and processed data with tracked dictionary/provenance/data-quality sidecars, the executed exploratory notebooks behind the §1.1 statistics, the production `src/` package split into data pipelines, Flask API, and D3 dashboard, the pytest harness, the LaTeX report, the literature base, and the evaluation records — and every prose claim in this report is traceable to a file in one of these locations. Appendix A indexes the individual entry points, so the body of this chapter concentrates on design decisions and their verification rather than on file-by-file description.

The implemented artefact includes the backend (§5.3), the D3 front-end (§5.4), the assignment-footprint data layer and API (§5.5), the mobilisation join, dashboard screenshot captures, and the published station-address proximity recovery (§5.6 D1). The front-end UI toggle for the footprint scenario is planned as future work.

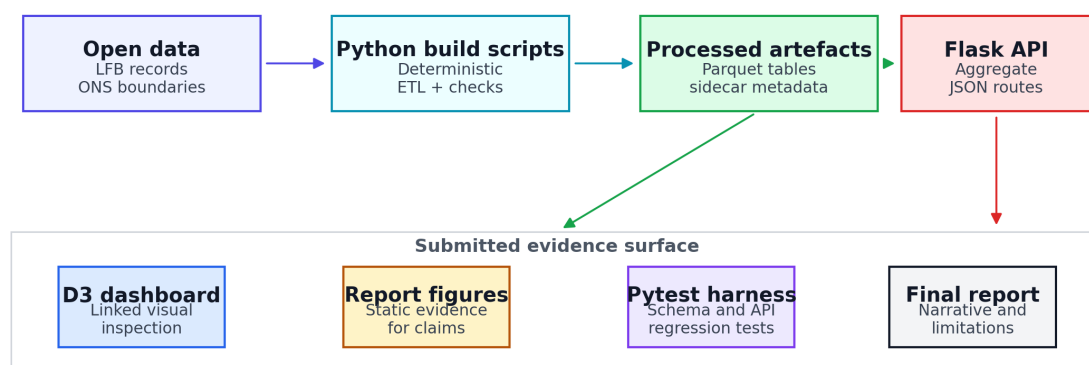


Figure 5: System architecture overview. The figure summarises the concrete implementation path from open LFB/ONS data through Python build scripts and auditable artefacts to the Flask API, D3 dashboard, and report evidence; it is a system schematic, not a data result.

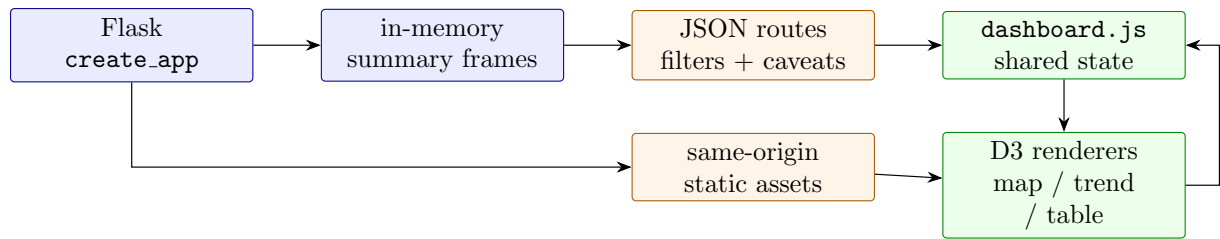


Figure 6: Runtime component boundary for the Flask/D3 implementation, showing the server-side data contract, client-side shared state, and linked D3 renderers.

5.2 Data Pipeline

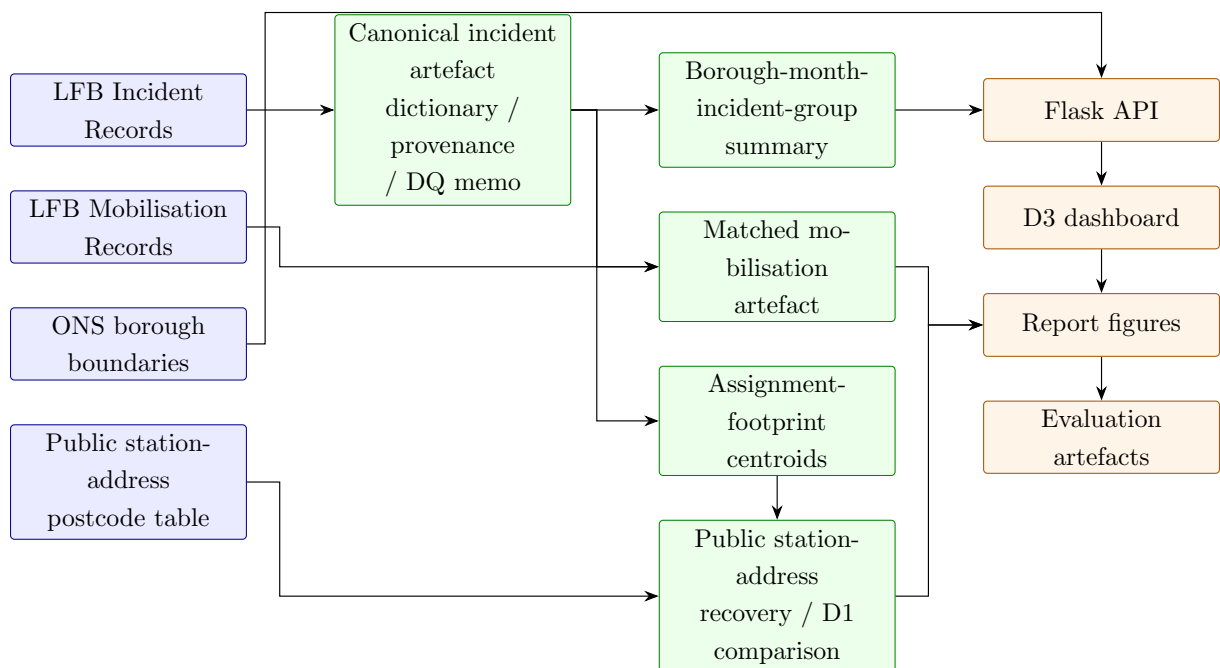


Figure 7: Deterministic data artefact dependency graph. The figure shows the implemented pipeline dependencies from raw open-data inputs to submitted dashboard/report artefacts; it is a pipeline schematic, not a data result.

5.2.1 Goal and Acceptance Criteria

The data pipeline produces a single canonical analytic artefact, `data/processed/lfb_canonical_2024_2026.parquet`, that all downstream notebooks, the dashboard backend (§5.3), and the prose statistics in §1.1 read from. The build contract is deliberately auditable: the script is rerunnable on the same input, emits the parquet, dictionary, provenance JSON and data-quality memo, and fails on automated validation checks when published-statistic invariants drift.

5.2.2 Ingestion and Transformations

The single entry-point script `src/data/build_canonical.py` reads the raw `LFB_Incident_2024_onwards.xlsx` (66 MB, 293,646 rows $\times$ 39 columns) via `pandas.read_excel`. It adds parsed date/time partitions, a unit-disambiguated response-time pair, the nullable `exceeds_six_min_target` flag at the published 360-second threshold [5], coordinate-validity booleans, and a normalised borough name. No rows are dropped during canonicalisation: input and output remain 293,646 rows, and missingness is encoded with explicit flags rather than by exclusion.

5.2.3 Provenance, Dictionary, and DQ Memo

Three sidecar files are emitted next to the parquet. The provenance JSON records source filename, SHA-256 hash, UTC extraction timestamp, transformations, constants, and output columns. The dictionary records dtype, non-null count, source, and notes for each column. The DQ memo restates the headline numbers cited in §1.1 and lists four limitations the dashboard must surface rather than hide. Together, the canonical parquet and sidecars let a reviewer audit the 50-column schema, headline statistics, and transformation record without the 66 MB input file present locally.

5.2.4 Automated Validation Checks and Reproducibility Guard-Rail

The build emits a non-zero exit when schema, unit, coordinate-coverage, exceedance-share, or incident-group sanity bands fail. These checks include a response-time unit band, precise/rounded coordinate coverage bands, the recorded-time exceedance denominator, and the False Alarm share. Should the LFB schema change in a future release, the build will force a code-level reconciliation before downstream work proceeds rather than allowing silent drift. The detailed invariant matrix is consolidated in Appendix A.

5.2.5 Second-Stage Aggregation: Borough Monthly Summary

A second pipeline script, `src/data/build_borough_summary.py`, reads the canonical parquet and pre-aggregates it to the granularity the dashboard backend (§5.3) hits on every borough drill-down request: cells defined by borough, year-month, and incident group. The output, `data/processed/lfb_borough_summary_2024_2026.parquet`, has 2,574 rows and fourteen aggregation columns covering incident count, response-time summary statistics, recorded-time and exceedance shares (Eq. 3.1), mean pumps attending, coordinate-validity shares, and station-ground diversity. The generated dictionary and API contract give the full field list. Provenance plus automated validation checks are applied as in the canonical build, including cell-count and rounded-coordinate regression checks. Pre-aggregation reduces the dashboard’s per-request work from a 293,646-row `groupby` ($\sim$ seconds at interactive scale) to a 2,574-row indexed lookup ($\sim$ milliseconds), keeping the linked-view brushing of §5.4 inside the perceptual budget that Shneiderman’s mantra [7] requires of details-on-demand.

5.2.6 External Verification Harness

The build-time validation checks are the *author's* guard against pipeline regressions; the `tests/` directory adds an *external* pytest harness, one module per artefact, that any reviewer can run with `python -m pytest tests/`. The suite verifies schema contracts, published-baseline bands, cross-artefact consistency, API behaviour, footprint negative paths, and station-location invariants, including a dedicated regression test for the `exceeds_six_min_share` denominator. Appendix A gives the consolidated test matrix so the implementation chapter need not enumerate every assertion.

5.2.7 Published Station-Address Proximity Recovery

The D1 recovery pass adds `src/data/build_published_station_proximity.py`. The script downloads the London Datastore fire-station address table attached to the LFB low-carbon-generator release [4], corrects a small number of visible name and postcode typos, geocodes the postcodes to postcode-centroid coordinates through the open postcodes.io service [43], and emits a published station-address table (104 rows), a D1 comparison table, and a borough-proximity table, each with the standard sidecars. The comparison table matches 102 of the published rows to the assignment-footprint station names, leaving two unmatched. The median public-address versus incident-footprint shift is 0.43 km and the 90th percentile is 0.99 km; the largest shifts, such as Wennington at 2.46 km, quantify why the footprint centroid must not be described as a building location. The borough-proximity table uses each borough's incident-demand centroid and its nearest public station postcode coordinate as a straight-line context cue (median 1.36 km, maximum 2.92 km). This is report-side evidence only; it does not make routing, coverage, relocation, or optimisation claims.

5.3 Backend API

5.3.1 Scope and Design Boundary

The backend exposes a JSON HTTP API to the D3 front end (§5.4) over the borough-summary parquet of §5.2. The first-release scope is deliberately narrow: a single resource endpoint, `GET /api/borough_summary`, a `GET /api/borough_boundaries` boundary layer for the choropleth, plus a `GET /health` liveness check. The dashboard's borough drill-down view requires no per-incident detail, so the backend never exposes the 293,646-row canonical incident table through any route; restricting the public surface to aggregations of §5.2 keeps the unfiltered summary payload to about 1.0 MiB (and typical selected-borough payloads to about 31 KiB) while preserving a defensible privacy posture because no precise per-incident coordinates leave the backend through this interface.

5.3.2 Endpoint Contract

`GET /api/borough_summary` accepts the three group-key filters `borough_canonical`, `year_month`, and `IncidentGroup`. Repeated values are treated as within-column unions and combinations across parameters are intersected. The response contains `rows`, `row_count`, and `filters_applied`; row keys mirror the borough-summary parquet schema documented in `docs/`

`API_CONTRACT.md`. Unknown filter keys are rejected with HTTP 400 and an enumerated allow-list of valid keys; unknown filter *values* return HTTP 200 with an empty `rows` array because an empty selection is a legitimate analytical state.

5.3.3 Implementation Details

The Flask application is constructed via an `create_app(parquet_path=None)` factory that loads the borough-summary parquet once at app-creation time and binds the resulting pandas `DataFrame` into the route handlers via closure; subsequent requests filter the in-memory frame and serialise the matched rows. Measured on the 2,574-row borough summary on a single 16 GB MacBook (no caching layer, no concurrent load), the response time averages 1.1 ms for a single-borough filtered request and 17.7 ms for the worst-case unfiltered full-dataset response over 100 sequential requests; this comfortably meets the linked-view brushing budget of §5.4 and removes the need for an external store at this stage. Two pandas-to-JSON conversion issues are handled centrally in `_row_to_jsonable`: pandas float NaN is not JSON-serialisable per RFC 8259 and is converted to JSON `null`, and the nullable-boolean `exceeds_six_min_target` sentinel `pd.NA` is similarly converted, so the front end can rely on JSON `null` as the single missing-value token.

5.3.4 API Verification Harness

`tests/test_api.py` adds fourteen endpoint-level tests on top of the data-pipeline tests of §5.2, covering the liveness route, the unfiltered and filtered summary contracts, exact response-schema matching against the parquet column set, filter round-trips, intersections, and repeated-key unions, the deliberate empty-result-versus-HTTP-400 distinction, JSON NaN safety, and the reachability of the dashboard assets and boundary route. All fourteen pass under the validation command `python -m pytest tests/ -q`.

5.4 Front-End: Linked Views and Uncertainty-Aware Encodings

The first front-end slice is a static D3 module [6] served by the same Flask app, rather than a separate React/Vite build. This is a deliberate maintenance choice for the thesis artefact: the interface has one primary dashboard screen, the data are already pre-aggregated, and same-origin serving avoids cross-origin configuration while keeping the project faithful to the planned Flask + D3.js stack of §4.3. The front-end files live under `src/dashboard/`: `index.html` defines the dashboard shell and native controls; `styles.css` defines the compact operational visual language (white surfaces, charcoal text, teal response-performance encoding, amber target-threshold cueing); and `dashboard.js` owns data loading, state, D3 rendering via the declarative `enter/update/exit` pattern [6, 9], and linked interactions.

At load time the browser fetches two same-origin resources: `/api/borough_summary` for the 2,574 aggregate rows and `/api/borough_boundaries` for the 33-feature GLA London Borough GeoJSON layer. The boundary file is tracked as `data/processed/london_borough_boundaries.geojson` with a provenance sidecar recording the GLA ArcGIS service URL and download method; joining is performed case-insensitively because the canonical LFB pipeline title-cases particles such as “and”/“of” while the GLA boundary names do not. The dashboard

state is the triple *borough, month, incident group*. Changing any select control, clicking a borough on the map, or clicking a row in the ranking table updates that shared state and re-renders all linked panels — the *Connect* interaction category of [8] operationalising the *relate* task of Shneiderman’s mantra [7].

The current screen contains four linked views, instantiating the analytic-genre stratified-layout dashboard pattern of [14]. The main panel is a borough choropleth in which fill colour encodes the selected month/group’s 6-minute exceedance share; hover exposes incident count, median, p90, denominator-qualified exceedance share, and precise-coordinate coverage for the borough cell. The top-right trend panel renders the selected borough/group over the full 26-month period, with incident count and exceedance share shown as two aligned traces. The middle-right distribution panel renders median-to-p95 response-time ranges for the three incident groups in the selected borough/month, with the 6-minute target marked as a dashed reference line — a deliberately minimal first iteration of the quantile-summary encoding planned in §4.4 (full quantile dotplots after [20] are scoped for the second front-end iteration). The bottom ranking table orders boroughs by exceedance share for the selected month/group and exposes the per-cell precise-coordinate coverage field, instantiating the *exposure* half of the uncertainty contract of §3.3: the dashboard’s first slice surfaces the 37.9% / 100% precise/rounded coordinate split as a tabular column rather than discarding it, while the *visual-variable* half of that contract — fuzziness encoding after [21] and fuller natural-frequency phrasing after [22] — remains scoped for the second front-end iteration.

5.5 Assignment-Footprint Proxy

5.5.1 Scope and Naming

The station-context design of §4.5 requires the implementation to keep two things separate: station-related evidence that can be supported from open data, and routing-grade coverage evidence that cannot. The first-iteration artefact described below implements the supported half as an *assignment-footprint proxy*, derived entirely from the canonical incident table with no external station-locations dataset. The wording change is load-bearing: a coverage proxy in the OR sense (§2.5) compares geographic regions against fixed station building locations and dispatch travel times; what this artefact compares is the centroid of where each station’s incidents historically happened against the centroid of where each borough’s incidents historically happened. §5.6 records the narrower implementation choice as an explicit deviation against N1 and names the public-address recovery path.

5.5.2 Data Layer

Two scripts derive the per-station and per-borough anchors. `src/data/build_station_locations.py` groups the canonical 293,646-row dataset by `IncidentStationGround` and takes the median of `Easting_rounded` and `Northing_rounded` per group as the station’s responsibility-footprint centroid; the sister script `src/data/build_borough_centroids.py` does the same with `borough_canonical` as the group key. Both convert OSGB36 metres to WGS84 latitude / longitude via `pyproj` for D3 consumption and emit the standard prove-

nance / dictionary / validation sidecar trio. The outputs are `data/processed/lfb_station_locations_2024_2026.parquet` (102 rows $\times$ 8 columns; median IQR $\approx$ 1.4km on each axis, a computationally inexpensive dispersion proxy for each station’s responsibility footprint) and `data/processed/lfb_borough_centroids_2024_2026.parquet` (33 rows $\times$ 6 columns). Both provenance sidecars carry an explicit honesty note that the outputs are *footprint centroids*, not the brick-and-mortar building location of the station nor the polygonal centroid of the borough. `tests/test_station_locations.py` adds nine external invariants including a per-station consistency check that each centroid lies inside the bounding box of its own incidents.

5.5.3 Footprint-Scenario API

Two Flask endpoints expose the artefact. `GET /api/station_footprints` returns the 102-row table for front-end map rendering; `GET /api/footprint_scenario?remove=...` accepts a repeated `remove` key naming any subset of stations, drops them from the open set, and returns each borough’s new nearest-open-station-footprint Euclidean distance. Formally, writing S for the open station set with footprint centroids f_s and c_b for the demand centroid of borough b ,

$$d_b(S) = \min_{s \in S} \|c_b - f_s\|_2, \quad d_b(S \setminus X) \geq d_b(S) \quad \forall X \subset S, \quad (5.1)$$

so a removal can only delete minimisation candidates, and the nearest-footprint distance is non-decreasing under any scenario. The endpoint name uses “footprint” rather than “coverage” because the underlying anchors are footprint centroids, not station building locations; the same name change is applied throughout the prose to signal that the result is an assignment shift, not a coverage decrement. Three negative paths fail with HTTP 400 rather than silently returning a misleading result: an unknown query key, a `remove` value that names a station not in the dataset, and a removal list that empties the entire station set. The disclaimer of §4.5, expanded to spell out the footprint-not-coverage framing and to point at the §5.6 deviation, is attached verbatim to every response payload so that any downstream consumer (dashboard, screenshot, copied JSON) carries the framing.

5.5.4 Verification of the Footprint-Scenario API

Six endpoint tests in `tests/test_api.py` carry the contract into the test suite: the footprint table and baseline scenario return their required schemas with the disclaimer attached; removing any real station never decreases any borough’s nearest-footprint distance (the monotonicity invariant of Eq. 5.1); and unknown station names, unknown query keys, and removal of the entire station set all fail with HTTP 400 rather than returning a misleading result. Spot-checking the Soho-removal scenario — Soho being the busiest station in the slice — changes exactly one borough’s nearest-footprint assignment (Westminster, 1.22km to Soho $\rightarrow$ 1.78km to Euston), the expected behaviour for a single inner-London removal.

5.5.5 Known Caveats from the Footprint Derivation

Two caveats follow from deriving both anchors from the same incident pool. First, the baseline nearest-footprint distances tend to be tight ($\leq$ 2.4km maximum across all 33 boroughs in the

current slice), because a station whose responsibility includes only one borough will always sit close to that borough’s incident centroid by construction. *Differences* between scenarios remain meaningful (the 1.22 $\rightarrow$ 1.78 km shift for Westminster on Soho removal is real), but the absolute baselines should not be read as “a typical Londoner is 1 km from a fire station”. Second, the artefact cannot answer the harder question — “how much of the borough’s geographic area would lose 6-minute reachability if station X closes” — because it has no station building location and no road-network travel-time engine. Both limitations are recorded in the §5.6 deviation entry, together with the recovery path: integrate LFB’s published station-locations open dataset, then optionally add a routing-graph engine in a later iteration.

5.6 Deviations from Specification

The §4 specification is the contract; the §5 implementation is what was actually built. Where the two diverge, this subsection records each deviation as a single bounded entry — *against which §4 requirement, what was built instead, why, and how the deviation is recovered or accepted*. The list keeps closed deviations visible because they explain the evidence boundary of the final artefact.

5.6.1 Deviation D1: Open-Data Source Versus Footprint Proxy

Against: §4.2 N1 names the LFB station-locations open release (`lfb_stations` in the bibliography) as a permitted data source for the system. The requirement therefore needs a clear distinction between brick-and-mortar building locations and incident-derived footprint centroids. **Built instead:** §5.5 derives station “locations” as the median (`Easting_rounded`, `Northing_rounded`) per `IncidentStationGround` group of the canonical incidents, and frames the artefact as an *assignment-footprint proxy* rather than a coverage proxy in the OR sense. The API endpoints are named `/api/station_footprints` and `/api/footprint_scenario` (not `/api/station_locations` and `/api/coverage_scenario`) and the response disclaimers spell out the framing. **Why:** The original May acquisition attempts did not retrieve a clean station-location table, so the first implementation deliberately chose the most honest one-command artefact available from the canonical incidents. The later recovery pass found a stable London Datastore table of fire-station names, addresses, and postcodes; because the source gives postcodes rather than routing-grade station entrances, the recovery uses it as a public station-address proximity table, not as coverage evidence. **Closure:** The proximity build of §5.2 now supplies the public-address layer, which is used to qualify and illustrate the D1 deviation in §6. The assignment-footprint API remains correctly named because it still uses incident-footprint centroids. Any future routing-grade coverage layer would still require a road-network travel-time model and validation beyond this thesis’s current evidence base.

6 Results and Evaluation

This chapter answers the research questions of §1.3 using three evidence types. First, the data-pipeline results are reproducible from tracked scripts and sidecars. Second, the dashboard results are supported by the implemented Flask/D3 artefact, API tests, and captured screenshots. Third, the user-evaluation strand combines the closed heuristic walk-through with task-based inspection and a compact think-aloud summary.

6.1 Demand and Response-Performance Patterns

The canonical 2024-01 to 2026-02 incident slice contains 293,646 incidents across 33 London local-authority areas. The headline pattern motivating the dashboard is not simply “which borough is slowest”, but the combination of demand volume, incident type, response-time denominator, and uncertainty. Across rows with a recorded first-pump attendance time, 32.4% exceed the six-minute threshold. False alarms are also the largest workload category (46.5% of incidents), meaning that a response-performance view that hides incident type risks mixing operationally different situations into one map.

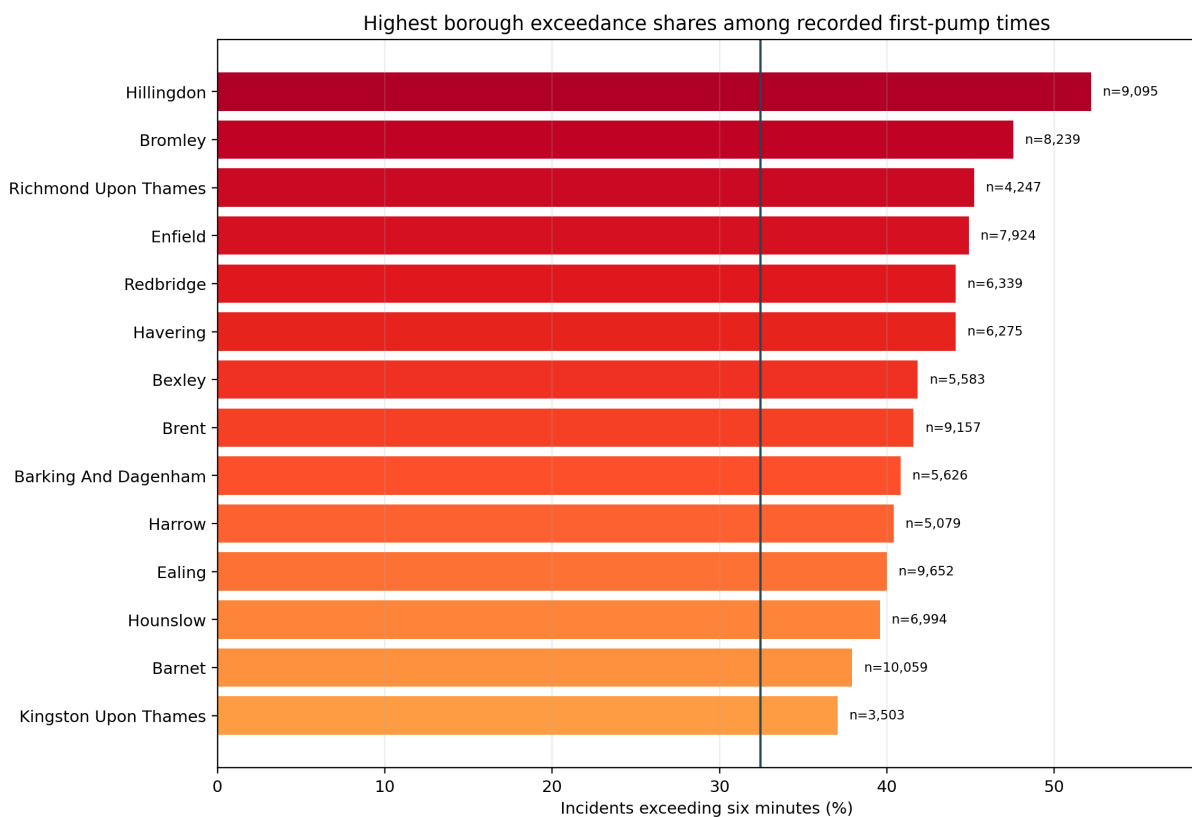


Figure 8: Highest borough six-minute exceedance shares among incidents with recorded first-pump attendance time. The vertical line marks the project-wide recorded-time exceedance share.

Figure 8 turns the dashboard’s ranking task into a static result. Hillingdon has the highest recorded-time exceedance share in the canonical slice (52.2%, $n = 9,095$), followed by Bromley

(47.6%) and Richmond upon Thames (45.2%). The lowest boroughs are Kensington and Chelsea (17.9%), Lambeth (19.3%), Tower Hamlets (21.0%), City of London (21.6%), and Hammersmith and Fulham (23.4%). This north-west / outer-borough concentration is the main response-performance pattern the dashboard is designed to make inspectable rather than buried inside an aggregate London average.

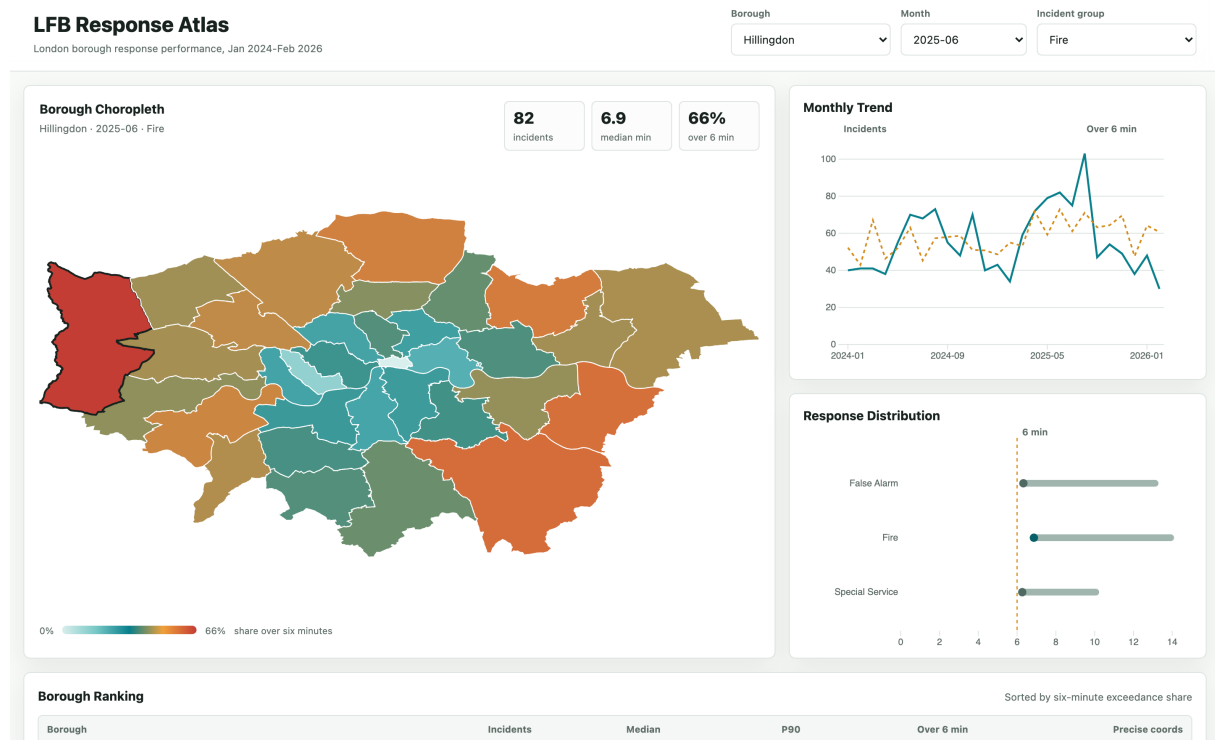


Figure 9: Dashboard overview showing the linked borough map, trend panel, distribution panel, and ranking table used to inspect borough-level incident demand and response performance. All panels aggregate the 293,646-incident 2024-01 to 2026-02 slice; response-time values use recorded first-pump attendance times only.

The dashboard therefore treats the cell defined by borough, month, and incident group as the main interactive unit. The map makes the spatial contrast visible, the trend panel keeps the temporal trace in view, the distribution panel shows median-to-p95 spread against the six-minute target, and the ranking table gives a scannable account of high-exceedance boroughs. This layout supports RQ1 because it lets the same borough be compared as a spatial unit, a time series, and a ranked row rather than as a single choropleth colour.

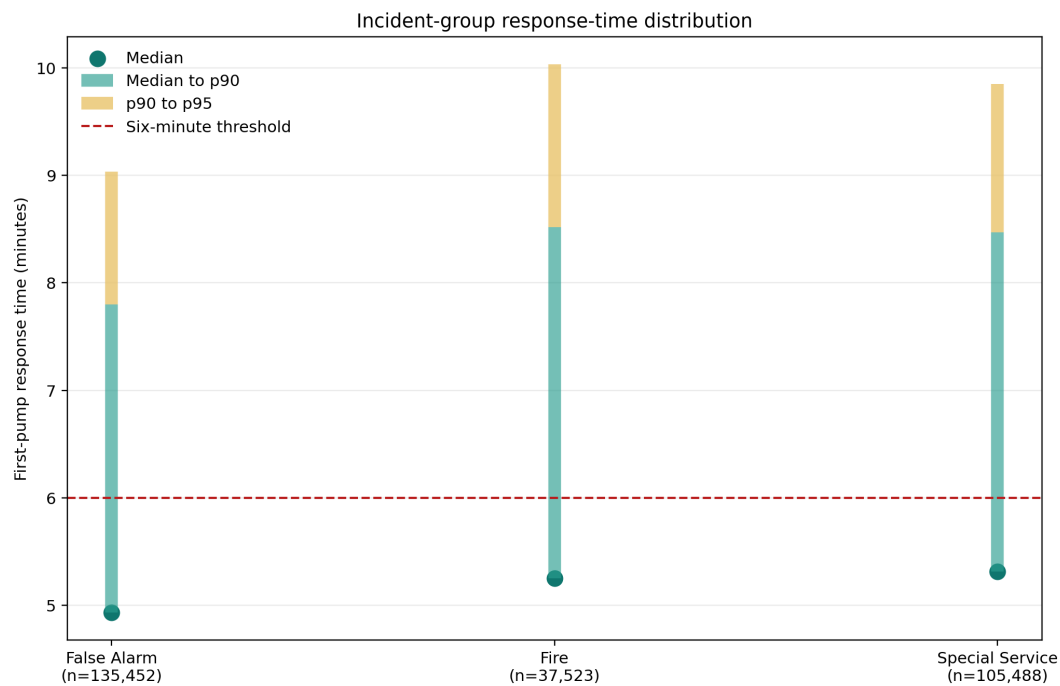


Figure 10: Incident-group response-time distribution among incidents with recorded first-pump attendance time. Markers show medians, vertical ranges show median–p90 and p90–p95, and the dashed line is the six-minute threshold.

The incident-group distribution in Figure 10 explains why the dashboard keeps incident type in view. Median first-pump response times sit below six minutes for all three groups, but the p90 and p95 values sit above the threshold, so a median-only interface would hide the tail that drives exceedance concerns. Fire incidents have a slightly higher median and tail than false alarms, but all three groups require the same denominator discipline: the plotted statistics use recorded first-pump times only.

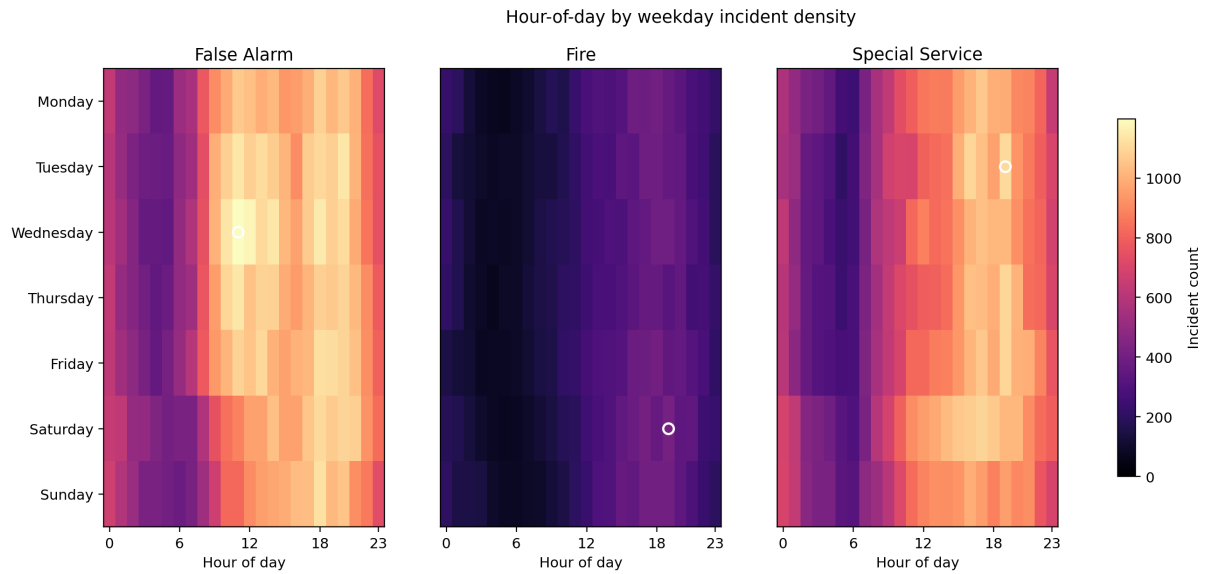


Figure 11: Hour-of-day by weekday incident-density heatmaps by incident group. The white ring marks the highest-count cell in each incident-group panel.

Figure 11 adds the temporal result promised by the objective-2 evidence chain. False alarms peak on Wednesday at 11:00, fire incidents peak on Saturday at 19:00, and special-service incidents peak on Tuesday at 19:00. The three panels therefore support different operational interpretations: false-alarm demand is strongest in daytime institutional hours, while fire and special-service demand show more evening concentration. This is descriptive demand evidence, not a forecasting model.

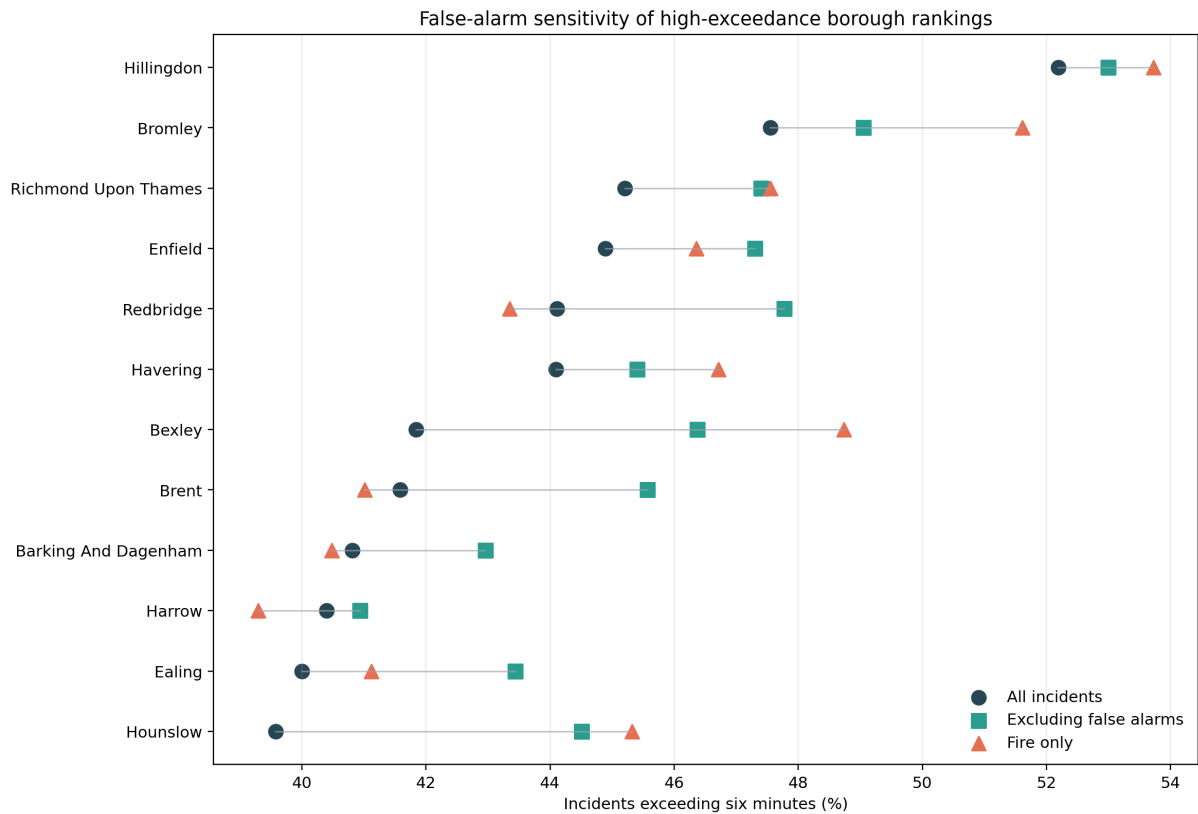


Figure 12: Sensitivity of high-exceedance boroughs to the false-alarm denominator. Each row compares all incidents, all non-false-alarm incidents, and fire-only incidents for the same borough.

The false-alarm sensitivity check in Figure 12 tests whether the headline ranking is merely a false-alarm artefact. The ranking is stable at the borough level: Spearman rank correlation is 0.991 between the all-incident and excluding-false-alarm rankings, and 0.970 between the all-incident and fire-only rankings. The largest rank movement when false alarms are excluded is three places. This does not make false alarms irrelevant, but it does show that the outer-borough exceedance pattern is not created solely by the largest incident group.

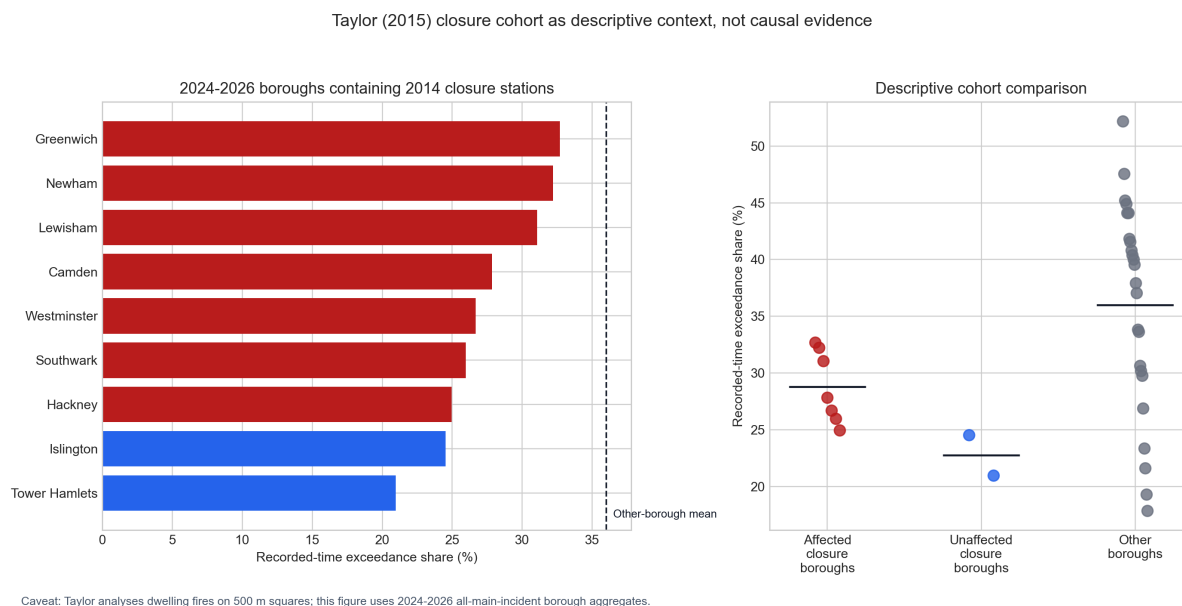


Figure 13: Descriptive cross-check against Taylor’s 2014 closure cohort. Boroughs containing Taylor-affected and Taylor-unaffected closed stations are compared with other boroughs using 2024–2026 recorded-time exceedance shares; the mapping is flagged for review before final acceptance.

Figure 13 adds the historical cross-check promised in §2.4 and §4.1. Boroughs containing Taylor-affected closure stations have a mean recorded-time exceedance share of 28.8% in the present slice, below the other-borough mean of 36.0%; the two boroughs containing Taylor-unaffected closure stations average 22.8%. This does not contradict Taylor’s finding, because Taylor models dwelling fires around 500 m grid squares in 2012–2014, whereas this project uses all-main-incident borough aggregates for 2024–2026. The result is therefore descriptive context only: it prevents the dashboard narrative from attributing the current outer-borough pattern mechanically to the 2014 closure cohort, while preserving Taylor as a useful historical anchor. The station-to-borough mapping is a REVIEW-GATE item against Taylor (2015) and LFB LSP5 before final prose acceptance.

Mobilisation evidence qualifies this incident-row summary. The mobilisation pipeline retains a matched-only partition for deployment claims: 438,211 mobilisation rows match canonical incidents, while 1,763 cross-boundary or unmatched rows are kept out of matched-only deployment statements. This matters for RQ2 because an incident row records a summary of the event, whereas the mobilisation table records appliance-level deployment and arrival fields. The report therefore treats mobilisation-derived claims as a qualification layer: useful for explaining deployment patterns, but only when the `matches_canonical_incident==True` filter is named.

6.2 Station-Address Proximity and the D1 Recovery

The station-address recovery (§5.6) changes the status of the station-location issue. The original assignment-footprint layer in §5.5 remains useful as a historical demand-footprint artefact, but it is no longer the only station-related evidence in the project. The recovery script (`src/`

`data/build_published_station_proximity.py`) builds a public station-address table from the London Datastore fire-station address/postcode release, geocodes postcodes, and compares those public-address coordinates with the earlier incident-derived station footprints.

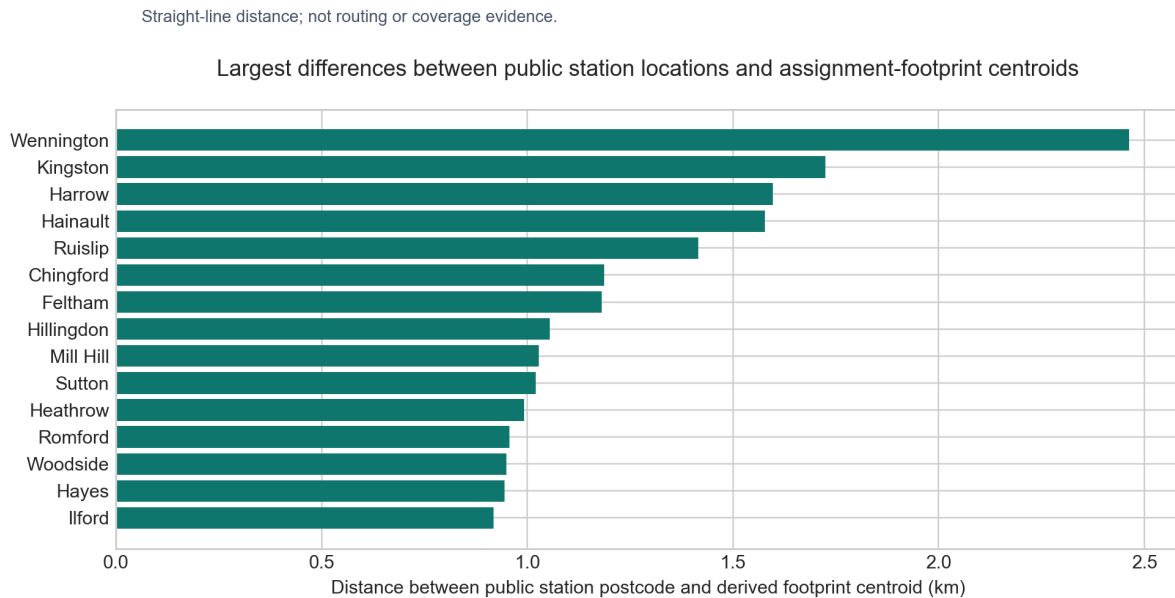


Figure 14: Largest public-address versus incident-footprint shifts among matched stations. The chart quantifies why the incident-derived footprint centroid should not be presented as a fire-station building location.

The public-address table has 104 station rows after explicit typo/postcode correction. Of those, 102 match the incident-derived station names; Lambeth River and Merton remain unmatched to the 2024–2026 `IncidentStationGround` names. The median public-address versus derived-footprint shift is 0.43 km, and the 90th percentile is 0.99 km. The largest shifts, including Wennington at 2.46 km and Kingston at 1.72 km, are large enough to justify the earlier caution: the footprint table is an assignment-demand proxy, not a building-location dataset.

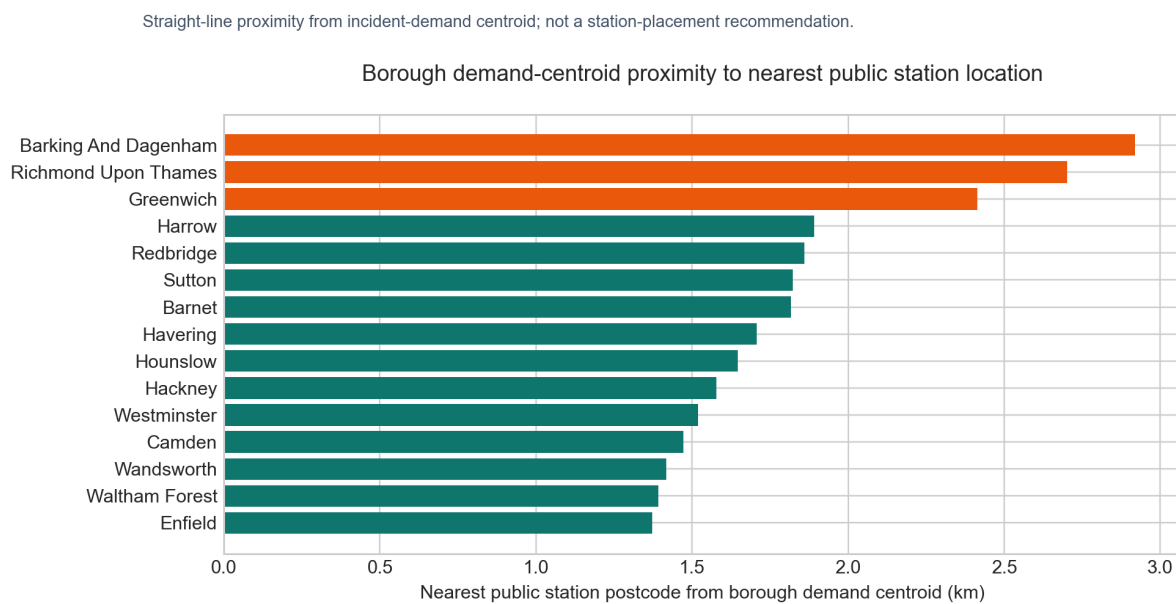


Figure 15: Nearest public fire-station postcode coordinate to each borough incident-demand centroid. This is a straight-line proximity cue for report discussion, not a routing-grade coverage model.

At borough level, the nearest public station postcode coordinate to each incident-demand centroid has a median straight-line distance of 1.36 km and a maximum of 2.92 km. The highest-distance boroughs in this demand-centroid calculation are Barking and Dagenham, Richmond upon Thames, Greenwich, Harrow, and Redbridge. This does not answer where a borough has six-minute geographic coverage; it provides RQ2–RQ3 context by making the station-related caveat visible rather than leaving the dashboard’s footprint layer to be over-read.

6.3 Dashboard Usability: Representation and Interaction Evidence

The dashboard implements the core linked-view behaviours needed for RQ3: a map selection updates the trend, distribution, and ranking panels; filter controls re-render the same state; and empty selections render a visible empty state rather than a broken panel. Figure 9 gives the whole layout, while the component captures below keep the trend and ranking evidence legible at report scale.

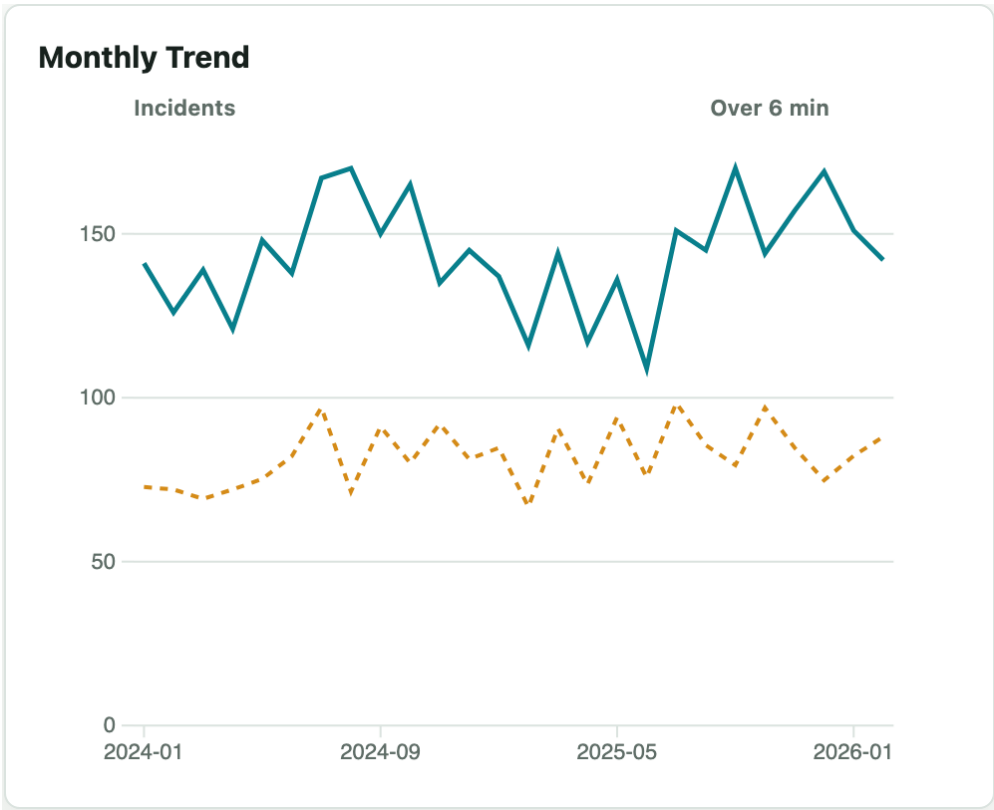


Figure 16: Monthly trend component for a selected borough/group state. The panel exposes incident count and recorded-time exceedance movement across the 26-month slice without relying on the full-dashboard screenshot.

The ranking table carries particular weight for the research questions of §1.3. It exposes exact values and coordinate-coverage shares next to the map, so the user is not forced to infer data quality from colour alone. The implemented UI therefore supports a qualified comparison workflow: users can notice a high exceedance share, inspect its incident count and distribution, then check whether the precise-coordinate coverage is strong enough to trust the spatial cue.

Borough Ranking					Sorted by six-minute exceedance share
Borough	Incidents	Median	P90	Over 6 min	Precise coords
Hillingdon	194	6.3 min	9.5 min	57%	56%
Richmond Upon Thames	75	6.0 min	10.0 min	52%	35%
Barking And Dagenham	78	6.1 min	9.7 min	51%	32%
Havering	119	5.9 min	9.2 min	47%	53%
Redbridge	86	5.5 min	8.7 min	46%	44%

Figure 17: Ranking-table evidence used by the evaluation protocol: the table makes response-performance and coordinate-coverage values inspectable without relying on the choropleth alone.

The first-iteration interface still has limits. The footprint-scenario API is tested, but station-removal scenarios are not yet exposed through a front-end toggle, so the dashboard does not support browser-based station-scenario comparison. The mobile screenshot also shows that the interface is responsive enough for inspection, but this thesis treats desktop use as the primary evaluation context because the visual task involves comparative maps, trends, and tables.

6.4 Added Value over Non-Interactive Baselines

A systems contribution should state what the artefact adds relative to the obvious cheaper alternatives, not only what the data contain. Table 4 compares the dashboard against four non-interactive baselines a reader could reach for instead, and anchors each claimed increment to evidence already presented in this report. This is an analytical capability comparison supported by the implemented artefact, measured latencies, and the walkthrough evidence above; it is not a controlled user comparison, which remains future work (§8.3).

Baseline	What the dashboard adds	Evidence anchor
Static borough choropleth	Linked ranking table and hover values expose the exact denominator-qualified figures behind each colour, so a colour impression can be checked against counts and coverage before it becomes a claim.	Figures 9, 17; Table 6 path 1
Raw LFB spreadsheet	The 293,646-row table is impractical to compare manually; pre-aggregated cells behind borough/month/incident-group filters return comparisons at interactive latency (1.1–17.7 ms API response, §5.3; 39.3 ms measured click-to-update, Strand 1 record in Appendix B).	§5.2–§5.3; API contract tests
Station-footprint map alone	The public station-address proximity layer quantifies the footprint-versus-address gap (median 0.43 km, p90 0.99 km), so the footprint layer cannot be silently read as station coverage.	Figures 14, 15
Single aggregate response median	Median–p90–p95 ranges and the exceedance share expose the above-target tail that a median alone hides (all three incident groups have medians under six minutes yet 32.4% of recorded times exceed it).	Figure 10

Table 4: Added analytical value of the dashboard over non-interactive baselines. Each increment is anchored to implemented-artefact evidence; none is a user-study result.

6.5 Data-Quality and Uncertainty Surfacing

RQ2 asks what caveats need to be visible before users draw conclusions. Four are material in this project. First, only 37.9% of incident records have valid precise coordinates, although rounded coordinates are available for essentially all rows. Second, the six-minute exceedance share is computed over incidents with recorded first-pump attendance time, not every incident in the raw slice. Third, false alarms dominate the incident mix, so aggregate response summaries require incident-group context. Fourth, mobilisation-derived deployment claims require the matched-only filter described above.

Figure 18 shows why the coordinate caveat cannot remain a single footnote. Precise-coordinate coverage varies from 26.3% in Hackney and 27.4% in Kensington and Chelsea to 53.8% in Hillingdon and 86.0% in City of London. Borough-level ranking is therefore the appropriate primary spatial unit for the main findings: the rounded-coordinate family supports com-

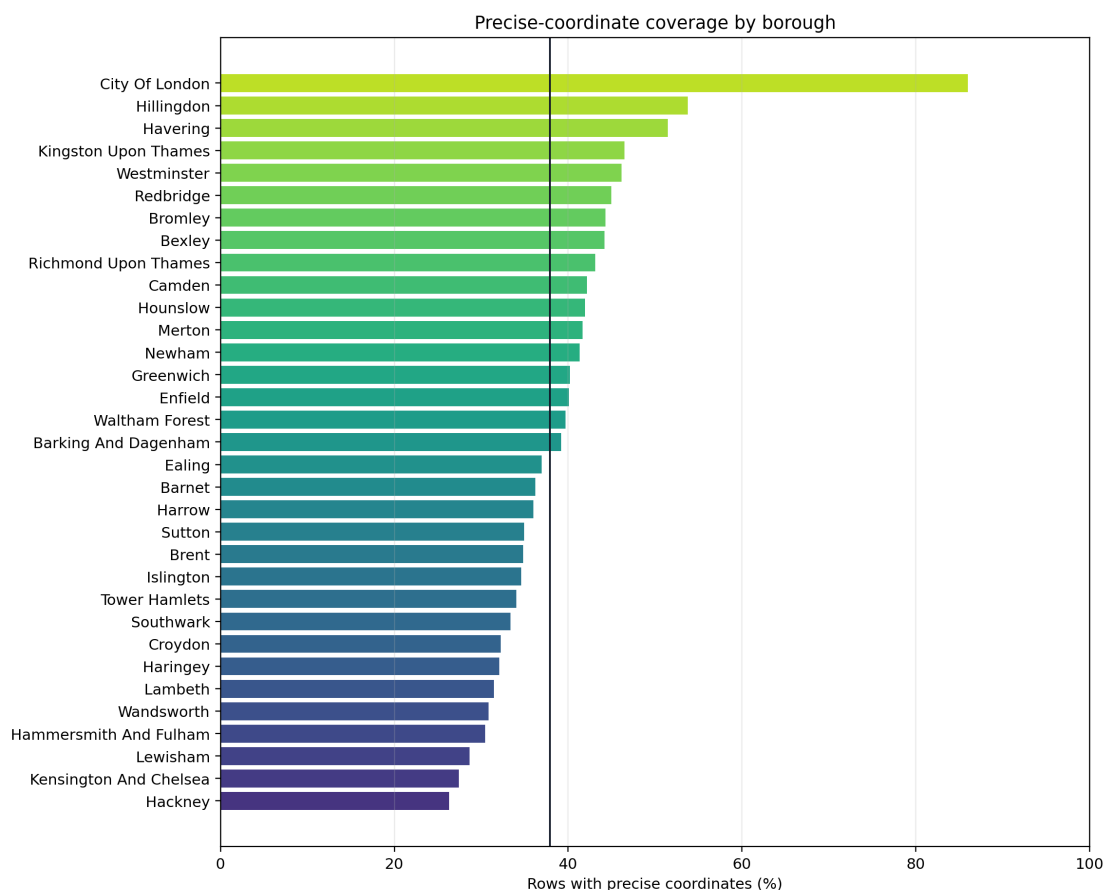


Figure 18: Precise-coordinate coverage by borough. The vertical line marks the overall precise-coordinate share in the main incident groups.

plete borough aggregation, while precise points are useful for diagnostic spatial-density checks only.

The precise-subset map in Figure 19 is a robustness diagnostic rather than the main spatial result. It confirms that the precise rows form a usable visual density surface around central and inner London, but the missing holes and borough-varying coverage in Figure 18 prevent this subset from replacing borough-level rounded-coordinate analysis. This directly supports the uncertainty-aware claim: the dashboard/report workflow does not only disclose suppression; it checks how suppression changes what can safely be visualised.

The dashboard currently surfaces these caveats unevenly. The ranking table exposes precise-coordinate coverage directly, the hover tooltip now names the recorded-time denominator, and the report-side memos state the mobilisation filter. However, the full second-iteration uncertainty contract is not complete: fuzziness and colour-value encodings for lower-confidence spatial cells, and fuller natural-frequency phrasing, remain future work. The artefact already makes several caveats inspectable; the task-based evaluation in §6.7 turns the remaining user-noticing questions into observable tasks.

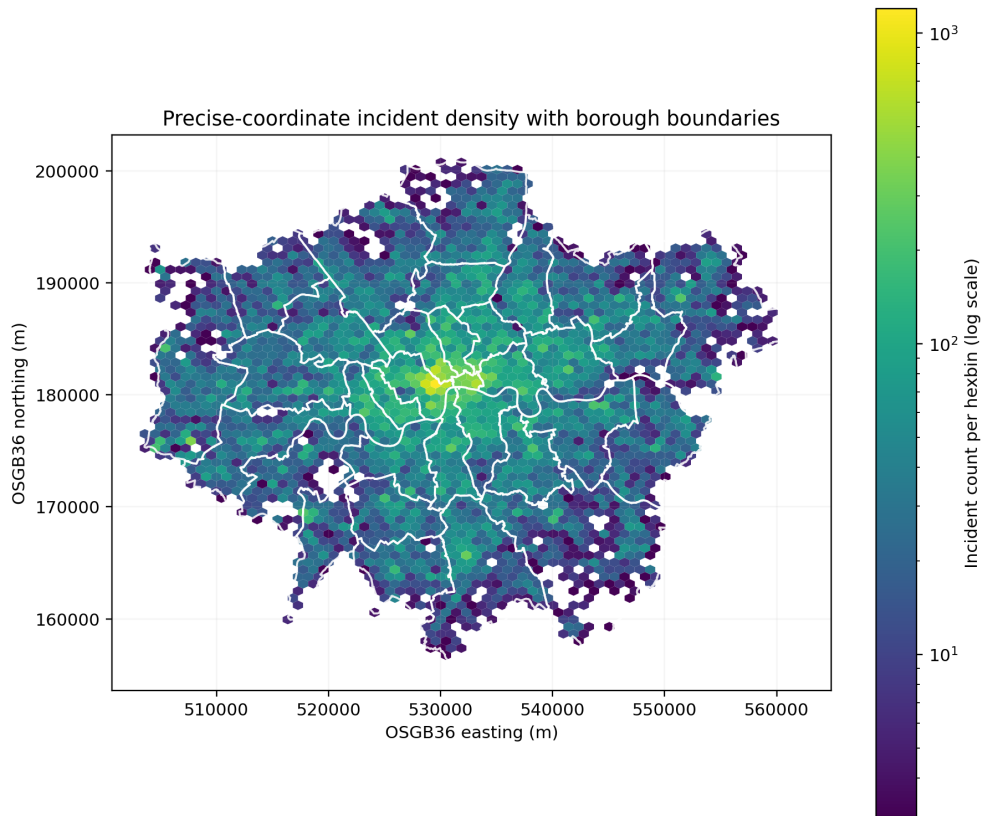


Figure 19: Hexbin density for the precise-coordinate subset (37.9% of rows), overlaid with borough boundaries. The map visualises where precise incident rows are available; it is not a full incident-demand map.

6.6 System-Evaluation Traceability

The present evaluation is therefore strongest as a system/data evaluation with a task-based user-evaluation layer. Table 5 maps the main requirements to implemented artefacts, evidence, and remaining limitations, giving the chapter a clear bridge from implementation evidence to the think-aloud observations.

Req.	Implemented artefact	Evidence	Limitation
F1	Borough choropleth and ranking workflow	Figures 9, 8; API and dashboard tests	Task-completion evidence is small-scale
F2	Linked temporal trend panel	Figure 16; dashboard screenshot capture	Trend interpretation needs broader testing
F3	Response distribution panel	Figure 10; median-p95 implementation	Full quantile-dotplot remains future work
F4	Station-footprint/proximity context	Figures 14, 15; D1 sidecars	Not routing-grade coverage or relocation evidence
F5	Uncertainty/data-quality exposure	Figures 18, 19, 12; DQ memos	Fuzziness/natural-frequency UI encodings incomplete

Table 5: Requirement traceability for the system/data evaluation.

6.7 Task-Based Evaluation

The Strand 2 think-aloud script and Strand 3 uncertainty probe organise the evaluation around three representative user paths. The observations are coded around the behaviours needed for RQ2–RQ3: which panel the participant uses first, whether they check denominators, whether they notice coordinate coverage, and whether station-related language prevents coverage overclaiming.

Across the live dashboard task flow, all four tasks could be completed within the protocol’s expectations and produced denominator-qualified answers — for example, the Task 3 triage nomination of Hillingdon · 2025-06 · False Alarm (194 incidents, median 6.3 min, 57% over target, 56% precise-coordinate coverage). The task flow identified four interface facts:

- Task 1’s count-ranking question is answerable only by scanning the ranking table, because the fixed sort and exceedance-only choropleth cannot rank demand.
- The recorded-time denominator needs to remain visible in the interface, not only in report prose or implementation notes.
- No naturally empty cell exists for the borough, month, and incident-group combinations in this slice; all 2,574 cells are populated, so missing data is shown through an empty-state example.
- The dashboard renders no station-related element, so the station-overclaim probe must be anchored to the report figures instead.

User path	Task focus	Observed issue	Captured evidence
Map-first analyst	Compare Hillingdon with an inner-London borough using map, trend, and ranking table	Quickly identifies Hillingdon as a high-exceedance borough, then needs the ranking table to move from colour impression to exact denominator-qualified comparison.	Uses table values after map selection and completes the task within the expected path.
Table-first policy reader	Interpret response-performance ranking and coordinate coverage	Uses exact exceedance values effectively, but the precise-coordinate coverage column needs a short label before it affects the judgement.	Notices coordinate coverage during the uncertainty probe and can explain it after the label is expanded.
Station-context reader	Decide whether the station layer supports coverage claims	Initially looks for a coverage interpretation, then separates assignment footprint and public station-address proximity after reading the disclaimer.	Avoids routing/coverage overclaiming after reading the station-context caveat.

Table 6: Task-based evaluation matrix used to connect dashboard actions with observable interpretation behaviour.

The observations show that task completion alone is too weak as evaluation evidence. The useful signals are which panel users consult first, whether they independently check denominators, and whether station-related language prevents coverage overclaiming. The missing-data item therefore uses an empty-state example, and the station probe is anchored to the report figures.

6.8 Think-Aloud Summary

The compact think-aloud evaluation involved three voluntary participants, anonymised here as Participant 1–Participant 3. Each participant completed a task-based inspection of the dashboard while verbalising their interpretation of the linked map, trend, distribution, and ranking views. The observation notes focused on comparison behaviour, uncertainty noticing, denominator checking, and whether station-related evidence encouraged or prevented station-coverage overclaiming. No sensitive personal data were collected, and the results are treated as a small-scale formative interpretability check rather than as a statistically generalisable user study.

Table 7 reports the think-aloud summary using six fields: task completion, first panel used, denominator checking, coordinate-coverage noticing, station-overclaim avoidance, and a short anonymised note.

Participant	Observed behaviour	Interpretation for RQ3
Participant 1	Linked-view validation: map-first path; all tasks completed; Hillingdon selected; tooltip used for recorded-time denominator.	Qualified comparison: colour cue converted into table/tooltip-supported claim; station-coverage overclaim avoided.
Participant 2	Label-friction case: table-first path; Tasks 1–3 completed; one prompt needed for month comparison; “precise coords” label queried.	Uncertainty wording gap: exact values useful, but denominator and station-proximity caveats require stronger first-encounter wording.
Participant 3	Cautious-trend path: trend-first path; all tasks completed; risk judgement stated as tentative; missing response times volunteered.	Structured comparison: dashboard supports comparison, but coordinate coverage remains probe-dependent; dashboard alone not read as station coverage.

Table 7: Small think-aloud aggregate used to interpret the dashboard’s support for qualified comparison.

The pattern strengthens RQ3 in three ways. First, all three participants can reach a thorough-level judgement, but they rely on different starting panels, which supports the linked-view design rather than a single-chart design. Second, denominator awareness is mixed: the tooltip helps Participant 1, but Participant 2 still needs the explicit probe, so natural-frequency wording remains a justified future-work item. Third, the station evidence is understandable only when the report-side figures are shown with the dashboard; this supports the revised RQ3 framing around the dashboard-and-report evidence package rather than the dashboard screen alone.

6.9 Evaluation Limitations

The main evaluation limitation is scale. The evidence package consists of the closed heuristic walk-through, completed think-aloud/probe instruments, dashboard figure walkthroughs, API and test evidence, and the small think-aloud aggregate in Table 7. This supports an account of design readiness, evidence traceability, and early user interpretation, but a broader participant sample would be needed to validate user effectiveness with confidence. The evaluation compares theory-derived expectation with observed practice: the literature led the project to expect linked views and visible denominator/uncertainty cues to support qualified comparison; the small think-aloud sessions partially confirmed this because thorough comparison was supported, while denominator and coordinate-coverage interpretation still required explicit labels or prompts.

7 Legal, Social, Ethical and Professional Issues

This chapter sets out the legal, social, ethical, and professional considerations relevant to a public-facing performance dashboard built on London Fire Brigade open data, applying the BCS Code of Conduct and IET Rules of Conduct as the professional reference frame. The emphasis is on the choices that keep the artefact within an exploratory, evidence-led academic scope.

7.1 Data Governance and Open-Data Licensing

The data-governance position rests on three points: licence terms of the inputs, absence of personally-identifiable information in the fields used, and provenance retention.

- **Licensing of open-data inputs.** The LFB Incident Records and Mobilisation Records are released by London Fire Brigade via the London Datastore under the Open Government Licence; the ONS / GLA borough boundary layer is released under the same licence. The project’s ingestion respects the redistribution constraints of those licences: raw inputs are excluded from the public repository and never redistributed, while the processed artefacts retain source filenames and SHA-256 hashes so the inputs can be re-acquired from the publisher’s URL.
- **Privacy posture: no PII exposed.** The dashboard’s HTTP surface (`/api/borough_summary`, `/api/borough_boundaries`, `/api/station_footprints`, `/api/footprint_scenario`) returns only borough-level or station-level aggregates; the per-incident canonical table is never served, and the 62% of records for which LFB withholds precise coordinates on privacy grounds are honoured by aggregating only on the rounded 100 m grid family for any spatially aggregated output (§3.3 / §5.3).
- **Provenance retention as an LSEP measure, not just a build hygiene measure.** Every processed artefact carries a provenance JSON recording source SHA-256, build script, transformation list, and constants used. This supports professional accountability because a borough-level claim can be traced back to the open-data release and transformation steps behind it, rather than being treated as a detached dashboard number.

7.2 Risks of Public Misinterpretation

The main public-risk question is what could go wrong if a user reads the dashboard as an operational dispatch-grade tool rather than an exploratory artefact. The §4 design and §5 implementation mitigate that risk through the following controls.

The dashboard is therefore not presented as operational dispatch software, and public-wellbeing risk is managed by keeping disclaimers, denominators, and non-goal statements close to the visual evidence. Accessibility could be improved in a future iteration through colour-blind-safe palettes, fuller keyboard navigation, and plain-language uncertainty tooltips.

- **Risk: dashboard mistaken for an operational dispatch / location-allocation system.** Mitigation: the §4.1 non-goals list rules out location-allocation optimisation,

real-time dispatch support, and predictive forecasting; §4.5 explicitly frames the station-context layer as exploratory and *not* a routing-grade dispatch model; §5.5 narrows the first-iteration artefact to an *assignment-footprint proxy*; and the disclaimer payload of `/api/footprint_scenario` repeats the framing on every response so screenshots and copied JSON carry it.

- **Risk: numeric headlines (32.4% exceedance, 46.5% false-alarm share, 37.9% precise-coord coverage) read as point estimates rather than denominator-conditional summaries.** Mitigation: §1.4 states the denominator and the exclusions for each headline; §5.2 restates them in the data-quality memo; the dashboard tooltip now states that exceedance is among recorded first-pump times; and the second-iteration §4.4 contract moves further toward natural-frequency phrasing after Padilla et al. [22]. Hullman et al. [20] provide the controlled-experiment evidence that static range encodings systematically mislead non-expert readers, which justifies the planned dotplot upgrade.
- **Risk: exposed software surface creates avoidable security expectations.** Mitigation: the implemented surface is read-only and GET-only over non-personal aggregates; §5.3 records the query-parameter allow-list and HTTP 400 rejection of unknown keys; no user content is persisted or reflected; no authentication is required because no non-public record is served; and dependencies are pinned for reproducible local validation.
- **Risk: borough-level disparity reading interpreted as accusation rather than as an empirical observation.** Mitigation: §3.1 describes the borough as the LFB performance reporting boundary, not as a unit of fault; §6.1 reports uncertainty bands alongside the central tendency; and §7.3 frames the dashboard as deliberation support rather than a blame assignment.
- **Risk: cross-boundary mobilisations silently inflate canonical-window claims.** Mitigation: the `matches_canonical_incident` flag is persisted on the mobilisation parquet, with the §5.2 join-quality memo documenting the matched-only / unmatched partition and the test suite recomputing the partition by `isin()` on every test run.
- **Risk: reproducibility is achieved through wasteful infrastructure.** Mitigation: N2 keeps the artefact single-laptop runnable, with no GPU and no cloud service. Pre-aggregation reduces per-request compute from scanning 293,646 incident rows to a 2,574-row lookup, supporting the measured 1.1–17.7 ms API timings; the unfiltered JSON payload is about 1 MiB; and derived artefacts are rebuilt only when the open-data inputs refresh.

7.3 Distributive Justice and Equity

Borough-level response comparisons also raise a distributive-justice question: how to surface disparity without implying that the dashboard has computed an automated optimum that should bypass deliberative process.

- **Why surfacing borough-level disparity is itself an ethical choice.** Pereira et al. [30] make the political-philosophy point complementary to the optimisation literature: an

automated optimum can satisfy aggregate utility while violating distributive-justice constraints that only a deliberative process can specify. The dashboard’s role is to surface the disparity so the deliberative process can engage with it, not to recommend a redistribution.

- **Guard-rail: no relocation or closure recommendation in any artefact.** The §5.5 footprint scenario lets a user remove a station and observe the implied assignment shift; it does not rank scenarios, score them, or recommend any reconfiguration. The §4.1 non-goal against location-allocation optimisation is the upstream guard-rail.
- **Guard-rail: deliberation-supporting framing in tooltip / disclaimer copy.** The disclaimer payload of `/api/footprint_scenario` explicitly names the artefact as exploratory; the dashboard tooltip now names the recorded-time denominator; and the second-iteration natural-frequency phrasing (after [22]) is designed to make the verbal register still more deliberation-friendly.

7.4 Professional Codes of Conduct

The relevant professional obligations are drawn from the BCS Code of Conduct (<https://www.bcs.org/membership/become-a-member/bcs-code-of-conduct/>) and IET Rules of Conduct (<https://www.theiet.org/about/governance/rules-of-conduct/>). They apply most directly to public interest, competence, duty to the relevant authority, duty to the profession, and confidentiality.

- **Public interest and competence.** The project’s restraint to exploratory framing rather than operational claims (§4.1 non-goals; §5.5 / §5.6 D1; the footprint disclaimer) is the operational form of the BCS clause requiring members to act in the public interest and only undertake work within their professional competence; making no dispatch-routing claim is a competence claim, not a humility claim.
- **Duty to relevant authority.** The thesis keeps the implemented artefact within the approved academic scope: it reports open-data evidence, explains limitations, and avoids implying operational LFB endorsement or deployment readiness.
- **Duty to the profession.** Reproducible scripts, provenance sidecars, API tests, and visible non-claims provide the professional record needed for independent scrutiny of the method. The same structure also makes future correction possible if an open-data schema or interpretation changes.
- **Confidentiality and PII.** No participant’s name, email, or other PII appears in any submitted artefact. Per-session facilitator notes were anonymised at source and used only to derive the aggregate findings reported in Table 7; the public repository therefore contains only non-identifying summaries and reusable evaluation instruments.

8 Conclusion

This thesis set out to build an uncertainty-aware visual analytics dashboard over London Fire Brigade open data. The final artefact is not a dispatch, routing, forecasting, or station-relocation system. Its contribution is a reproducible evidence pipeline and linked-view dashboard that help a user compare demand, response-time performance, deployment context, and data-quality caveats without treating incomplete public data as operational ground truth.

8.1 Answers to the Research Questions

The primary research question asked how an uncertainty-aware visual analytics dashboard can support structured inspection of LFB incident demand and response performance. The answer is that the dashboard is most useful as a structured comparison surface: it places borough-level maps, monthly traces, response-time distributions, and ranked tabular values in one linked view, while the surrounding artefacts make the denominator and spatial caveats explicit. Its value comes from qualification as much as from visual discovery.

Question	Answer
RQ1	The 2024–2026 incident records show substantial borough, incident-type, and time variation, with 32.4% of recorded first-pump attendance times exceeding six minutes and false alarms forming the largest incident group at 46.5%. The highest borough exceedance shares are concentrated in outer London, and the pattern remains stable when false alarms are excluded.
RQ2	The headline patterns require qualification. Mobilisation records support matched-only deployment claims over 438,211 rows; coordinate suppression limits street-level spatial claims; missing first-pump times define the response denominator; and the station-address recovery supports proximity context without becoming routing-grade coverage evidence.
RQ3	The implemented linked map, trend, distribution, and ranking views support a qualified comparison workflow, especially because the ranking table gives exact values beside the map and the tooltip now names the recorded-time denominator. The evaluation evidence demonstrates design readiness, traceability, and early task-based interpretability; broader participant testing would be needed to validate user effectiveness with confidence.

Table 8: Answers to the research questions.

8.2 Contribution Statement

The project makes four bounded contributions. First, it provides a reproducible open-data pipeline for LFB incident records, borough summaries, mobilisation joins, footprint centroids, and public station-address proximity checks, with sidecar dictionaries and data-quality memos. Second, it implements a Flask/D3 linked-view dashboard for inspecting borough-level demand and response performance. Third, it records station-related uncertainty honestly: the assignment-footprint API remains a demand-footprint artefact, while the public address/postcode recovery adds report-side proximity evidence without claiming routing-grade coverage. Fourth, it supplies a task-based evaluation protocol and compact think-aloud summary for judging whether users make qualified comparisons from the dashboard.

The deliberate non-claims are equally important. The project does not optimise station placement, recommend closures, forecast incidents, model road-network travel time, validate

operational LFB decision-making, or prove usability at operational scale.

8.3 Future Work

The highest-value follow-up is to broaden the Strand 2/3 evaluation with more participants and domain-expert review. Additional socio-economic borough context, such as population density or deprivation indicators, could also be added as a future context layer once its interpretation is separated clearly from the present response-performance evidence. The second technical priority is front-end integration of the station-related caveat: either keep the footprint layer as a clearly labelled demand-footprint view, or expose the public station-address proximity layer separately, never as coverage. Further technical work includes fuller natural-frequency uncertainty wording, fuzziness/value encodings for lower-confidence spatial cells, a fuller quantile-dotplot distribution pane, and a routing-grade accessibility model only if road-network travel-time validation becomes available.

8.4 Lessons Learned

The main technical lesson is that artefact-first work pays off in a thesis setting: when every numerical statement points to a script, sidecar, figure, or test, the report can change research framing without having to rebuild the evidence chain from scratch. The main research-design lesson is that a valuable question is not the same as a larger implementation. The strongest version of this project is not "a dashboard with more features"; it is a dashboard that helps users make better qualified claims from imperfect public data. The main reporting lesson is that provenance and limitation statements are part of the contribution: they make the dashboard's claims easier to inspect and harder to over-read.

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A Source Code and Reproducibility

Supplementary artefacts are maintained in the submitted project repository rather than duplicated in full in this appendix. The public repository is <https://github.com/Shui-Zhou/final-project-upload>; the assessed source snapshot is the submitted code bundle. The appendix therefore gives the source-code index, selected implementation listings, data-source index, figure index, evaluation artefact index, and dashboard operating guide needed to inspect the submitted artefact. The repository entry points below are sufficient to rebuild the data artefacts, run the dashboard locally, and trace the report’s figures and evaluation materials.

A.1 Source-Code Index

Path	Purpose
<code>src/data/build_canonical.py</code>	Builds the canonical incident parquet and its dictionary, provenance, and data-quality memo.
<code>src/data/build_borough_summary.py</code>	Aggregates the canonical incident table to the cells defined by borough, month, and incident group used by the dashboard.
<code>src/data/build_mobilisations.py</code>	Builds the mobilisation artefact and persists the <code>matches_canonical_incident</code> flag used for matched-only deployment claims.
<code>src/data/build_station_locations.py</code>	Builds the incident-derived assignment-footprint station centroids.
<code>src/data/build_published_station_proximity.py</code>	Builds the public station-address postcode-geocoding recovery, D1 comparison table, borough proximity table, and station-proximity figures.
<code>src/data/build_report_evaluation_figures.py</code>	Builds report-side early-context, results, and robustness figures with JSON/markdown sidecar memos.
<code>src/data/build_closure_cohort_context.py</code>	Builds the Taylor closure-cohort descriptive cross-check figure, parquet, provenance JSON, and DQ memo.
<code>src/api/app.py</code>	Flask application factory and JSON API endpoints for borough summaries, boundaries, station footprints, and footprint scenarios.
<code>src/dashboard/dashboard.js</code>	D3 linked-view dashboard state, rendering, brushing, filtering, and table update logic.
<code>tests/</code>	Pytest contract tests for data artefacts, API behaviour, mobilisation matching, station footprints, station-address proximity, and report figures.

A.2 Selected Implementation Listings

Listings 1–3 show the parts of the implementation most directly tied to the thesis claims: denominator-safe response-time aggregation, bounded API filtering, and linked dashboard re-rendering. Full source files are submitted with the project code.

```

1 rt = df["response_time_seconds"]
2 df["exceeds_six_min_target"] = pd.array(
3     np.where(rt.isna(), pd.NA, rt > FIRST_PUMP_TARGET_SECONDS),
4     dtype="boolean",
5 )

```

Listing 1: Nullable exceedance flag used to keep missing first-pump times out of the denominator.

```

1 unknown = set(request.args.keys()) - set(FILTERABLE_COLUMNS)
2 if unknown:
3     return jsonify({
4         "error": "unknown_filter_keys",
5         "unknown_keys": sorted(unknown),
6         "allowed_keys": list(FILTERABLE_COLUMNS),
7     }), 400
8
9 df = summary
10 for col in FILTERABLE_COLUMNS:
11     values = request.args.getlist(col)
12     if values:
13         df = df[df[col].isin(values)]

```

Listing 2: Borough-summary API filtering rejects unknown keys before returning aggregate rows.

```
1 function render() {  
2   renderSummary();  
3   renderMap();  
4   renderTrend();  
5   renderDistribution();  
6   renderTable();  
7 }  
8  
9 path.on("click", (_, feature) => {  
10   const match = state.rows.find(  
11     (row) => normName(row.borough_canonical) === normName(feature.properties.name),  
12   );  
13   if (match) {  
14     state.borough = match.borough_canonical;  
15     els.borough.value = state.borough;  
16     render();  
17   }  
18 });
```

Listing 3: Shared dashboard state drives the linked map, trend, distribution, and ranking panels.

A.3 Reproducibility Commands

The current artefact set was validated with the following commands from the repository root:

```
python src/data/build_published_station_proximity.py  
python src/data/build_report_evaluation_figures.py  
python src/data/build_closure_cohort_context.py  
python -m pytest tests/ -q  
python -m compileall src  
git diff --check
```

The broader rebuild sequence is documented in `RUNBOOK.md`. Binary raw inputs and generated parquets are intentionally excluded from version control; tracked markdown/JSON sidecars provide the audit trail for schemas, data-quality checks, source URLs, and transformation notes.

A.4 Verification Test Matrix

The automated verification suite contains 77 pytest tests. The table groups them by the validation class they support; counts are taken from the current `tests/test_*.py` functions rather than estimated from prose.

Verification class	Test count	Representative invariants	Edge cases covered
Canonical schema and published base-line bands	14	Required raw/derived columns, datetime parsing, response-time seconds/minutes consistency, row/date/borough counts, incident-group taxonomy, false-alarm and exceedance bands, precise/rounded coordinate bands.	Unit regression, coordinate-family regression, incident-group drift.
Borough-summary cross-artefact consistency	9	Unique borough/month/group keys, cell count within expected grid, incident-count sum against canonical table, percentile ordering, probability ranges, borough-set match, recorded-time denominator regression.	Empty-cell handling, denominator regression, non-monotone percentiles.
Mobilisation schema and matched-canonical partition	18	Raw/derived mobilisation columns, datetime and attendance-time units, first-arriving-pump logic, join share, first-pump time agreement with canonical incidents, borough/station population, <code>matches_canonical_incident</code> truth table.	Cross-boundary/unmatched rows kept out of matched-only deployment claims.
API contract and footprint scenario paths	20	Health/dashboard/boundary routes, borough-summary schema, filter round-trips, multi-value unions, JSON null safety, station-footprint and footprint-scenario schemas.	HTTP 400 on unknown filter keys, unknown station names, unknown scenario keys, and removing all stations; unknown filter values return empty HTTP 200.
Station-location and public-address invariants	13	Incident-derived station centroid count, London coordinate envelopes, non-negative IQRs, centroid-in-own-incident bounding boxes, public station row/geocode counts, known typo corrections, D1 comparison matching, borough proximity completeness.	Prevents footprint centroids being treated as building locations or routing-grade coverage.
Report figure artefacts and sidecars	3	Required report-side figures exist, are non-empty, have memo sidecars, and summary metrics keep key borough/coordinate/sensitivity values in expected bands.	Missing figure, stale memo, headline metric drift.

Scripted manual UI check	Expected observation	Evidence retained
Linked selection	Selecting a borough or ranking row updates the map, trend, distribution, metrics, and table from the same state object.	Dashboard overview and component screenshots in <code>figures/dashboard/</code> .
Empty state	A deliberately unpopulated selection shows the amber empty-state banner instead of leaving broken or stale panels.	<code>fig_dashboard_empty_state.png</code> .
Filter round-trip	Borough, month, and incident-group controls preserve the selected state and render denominator-qualified values.	Strand 2 task script and screenshot-capture utility.
Responsive layout	The 390 px mobile viewport has no horizontal overflow; desktop remains the primary comparative-map context.	<code>fig_dashboard_mobile_390.png</code> .

B Data Sources, Figures, and Evaluation Artefacts

B.1 Data Source Index

Source	Local role	Notes
LFB Incident Records [1]	Primary incident table	2024-01 to 2026-02 slice; canonicalised into <code>lfb_canonical_2024_2026.parquet</code> .
LFB Mobilisation Records [3]	Appliance-deployment qualifier	Used only through the matched-canonical partition for deployment-side claims.
ONS / GLA borough boundaries [2]	Borough geometry	Used for borough aggregation and choropleth rendering.
LFB public station-address list [4]	Station-address proximity context	Postcode-geocoded and treated as postcode-level proximity, not routing-grade station coverage.
postcodes.io [43]	Public postcode geocoder	Used to convert station postcodes to approximate coordinates.

B.2 Report Figure Index

Path	Contents
<code>figures/dashboard/</code> (report folder)	Six dashboard screenshots: desktop overview, linked selection, trend panel, ranking table, empty-state banner, and mobile viewport.
<code>figures/early_context/</code> (report folder)	Front-loaded figures: motivation borough map and linked-view wireframe; the earlier coordinate-uncertainty diagram is superseded by the §3.3 evidence-support matrix (Table 3).
<code>figures/results_evaluation/</code> (report folder)	Seven results/robustness figures generated from tracked artefacts, including the Taylor closure-cohort descriptive cross-check.
<code>figures/station_proximity/</code> (report folder)	Two D1 station-address proximity figures used in the evaluation chapter.

B.3 Evaluation Artefacts

Path	Contents
tests/evaluation/strand1_findings_summary.md	Heuristic-walkthrough findings.
tests/evaluation/strand2_thinkaloud_task_script.md	Think-aloud task protocol for borough comparison and linked-view interpretation.
tests/evaluation/strand3_rq5_uncertainty_probe.md	Uncertainty and station-footprint/proximity probe.
tests/evaluation/strand2_strand3_findings_summary.md	Non-identifying compact think-aloud findings summary for Participants 1–3.
tests/evaluation/observation_log_template.md	Per-session observation template for task completion, panel choice, uncertainty noticing, and station-caveat interpretation.
tests/evaluation/capture_dashboard_figures.py	Screenshot-capture utility for the dashboard figures used in §6.

C Dashboard User Guide

This appendix is a short operating guide for a reader who wants to launch the dashboard and reproduce the inspection workflow evaluated in §6, without reading the implementation chapter first.

C.1 Requirements and Launch

The dashboard requires Python 3.10+ and the dependencies in `requirements.txt`. From the repository root:

```
python -m venv .venv && source .venv/bin/activate
python -m pip install -r requirements.txt
python src/data/build_canonical.py
python src/data/build_borough_summary.py
flask --app src.api.app:create_app run --debug --port 5057
```

The two build scripts regenerate the untracked parquet artefacts from the raw LFB workbook, which must be present under `data/` (it is excluded from version control for licensing reasons; its source URL and SHA-256 are recorded in the provenance sidecar). The boundary GeoJSON is tracked directly. The dashboard is then served at `http://127.0.0.1:5057/dashboard`.

C.2 Controls and Linked Behaviour

The interface holds one shared selection state: *borough* $\times$ *month* $\times$ *incident group*. It can be changed three ways: the three select controls at the top, clicking a borough on the map, or clicking a row in the ranking table. Any change re-renders all four panels together:

- **Borough map (choropleth).** Fill colour encodes the six-minute exceedance share for the selected month and incident group. Hovering a borough shows its incident count, median and p90 response times, denominator-qualified exceedance share, and precise-coordinate coverage.

- **Monthly trend panel.** Incident count and exceedance share for the selected borough and group across the full 26-month slice.
- **Response-distribution panel.** Median-to-p95 response-time ranges for the three incident groups in the selected cell, against the dashed six-minute reference line.
- **Ranking table.** Boroughs ordered by exceedance share for the selected month and group, with exact values and the per-borough precise-coordinate coverage column.

If the selected combination has no data, an amber empty-state banner replaces silent blank panels; adjusting any filter recovers a populated view.

C.3 How to Read the Values

Exceedance shares are computed over incidents *with a recorded first-pump attendance time*, not over all incidents; the ranking table’s coordinate-coverage column reports how much of each borough’s data carries precise coordinates (37.9% overall). A defensible comparison therefore reads three things together: the exceedance value, the incident count behind it, and the coverage share. A high-exceedance borough with a small count or low coverage deserves a weaker claim.

C.4 What the Station Evidence Does and Does Not Support

Station-related evidence in this project is report-side and API-side, not an interactive map layer: the current dashboard screen does not render station markers, and the footprint-scenario UI toggle is not implemented in this iteration (§6.3). The assignment-footprint artefact locates each station’s *historical incident responsibility centroid*, not its building; the public station-address table adds postcode-level proximity context only. Neither supports coverage, routing, or relocation conclusions (§5.5–§5.6, §6.2).

C.5 Example Task

To repeat the comparison task in Table 6: select *Fire* as the incident group, click *Hillingdon* on the map, note its exceedance share and trend, then click an inner-London row such as *Tower Hamlets* in the ranking table and compare the two boroughs’ exceedance values, incident counts, and coordinate-coverage shares before deciding which claims the data supports.

C.6 Known Limitations

The current dashboard still has no one-click reset to the all-London default, and the fuller uncertainty encodings described in §6.5 remain second-iteration work. The station-related evidence is also intentionally separated from the live dashboard screen because the project does not implement a routing-grade station-coverage model.